



GE VERNOVA

ADMS

16.25.1

OMS Read-Only Interface Programmer Guide

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1. Introduction

This document describes the Outage Management web server interface (Net Data Server or OMS Data Service) that enables read-only access to OMS data.

Net Data Server supplies data for cleared and canceled incidents to the History Correction and Data Correction grid tabular displays of the OMS Display plugin, for reviewing and editing historical data.

Using the Net Data Server interface, customers may provide a web-based presentation of ADMS information both inside and outside of the control center.

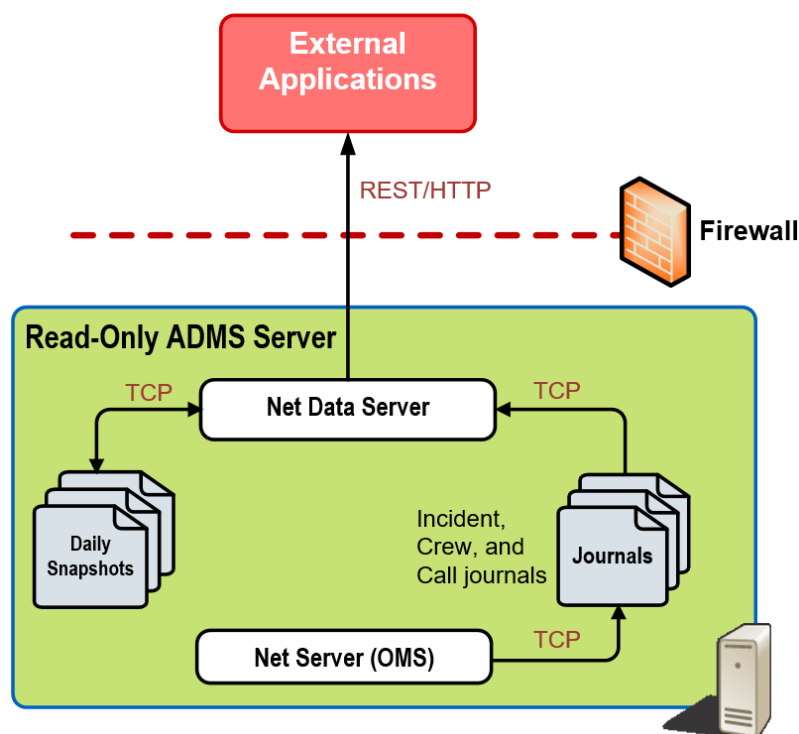


Figure 1. Figure Outage Management Interface

The document describes the approach, architecture, query types, and data collections that are supported by the Net Data Server web service.

Net Data Server web service is exposed using the Windows Communication Foundation (WCF) Data Services, which is Microsoft's standard framework for web services. Thereby it is recommended to use Microsoft's solutions for providing security.

Net Data Server web server uses the Representational State Transfer (REST) architecture and is Service-Oriented Architecture (SOA) compliant. The security aspects include using a firewall and an HTTPS connection.

There are two options for hosting Net Data Server web services in ADMS. The self-hosted option, where Net Data Server exposes the web service in a Windows form, is shown in figure below.

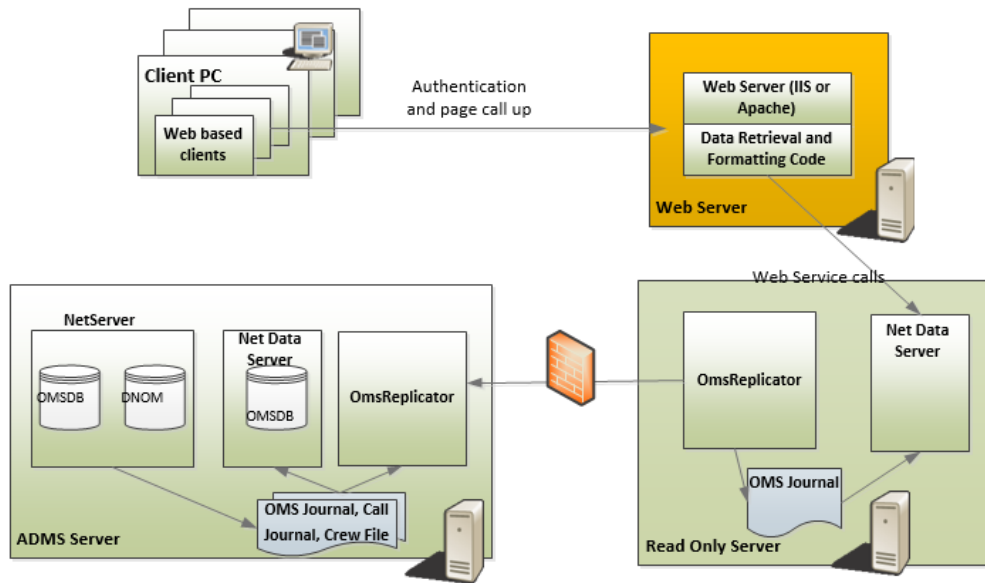


Figure 2. Figure Self Hosted Service

The second option, where the Net Data Server's OmsDataService is hosted directly in IIS, is shown in figure below.

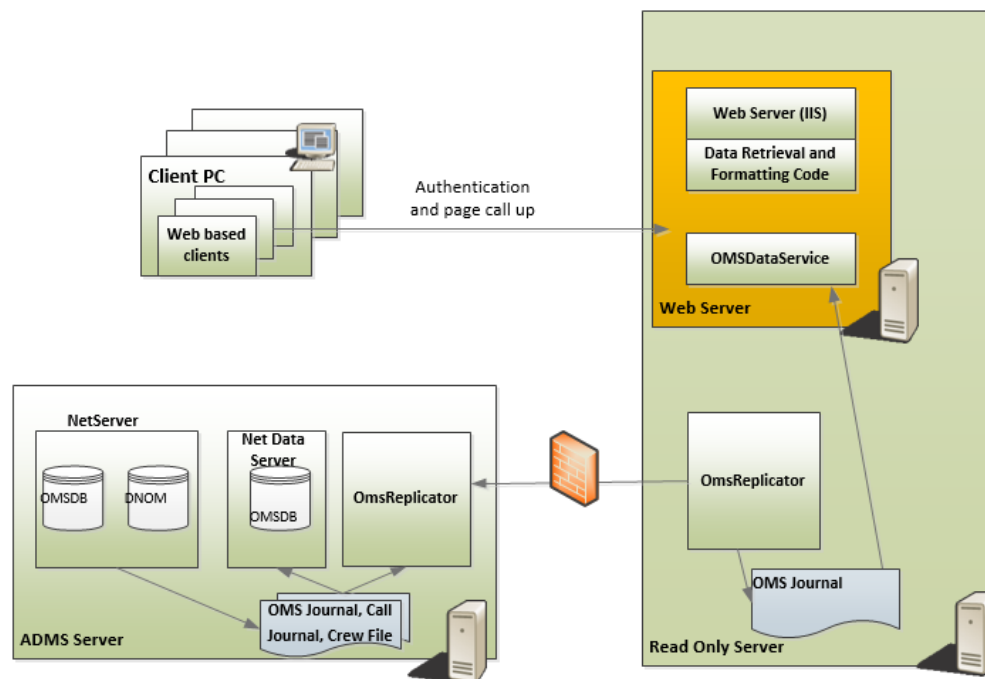


Figure 3. Figure Service Hosted in IIS

2. Web Service Queries

Resources are exposed based on conventions defined in the ODATA protocol. Details about ODATA can be found at <http://www.odata.org>. This specification allows queried data to be returned in AtomPub or JSON format. The default format to return data is AtomPub. GE uses the WCF Data Services framework to expose the services. Details about WCF data services are available at <http://msdn.microsoft.com/en-us/data/bb931106>.

Navigating to the home URI "http://localhost:8800/Oms" (configurable) for the web service lists all exposed resources as shown below.

Web Service Collections

```
<?xml version="1.0" encoding="utf-8" ?>
<service xml:base="http://localhost:8800/Oms/"
  xmlns="http://www.w3.org/2007/app"
  xmlns:atom="http://www.w3.org/2005/Atom">
  <workspace>
    <atom:title>Default</atom:title>
    <collection href="Incidents">
      <atom:title>Incidents</atom:title>
    </collection>
    <collection href="Crews">
      <atom:title>Crews</atom:title>
    </collection>
    <collection href="Employees">
      <atom:title>Employees</atom:title>
    </collection>
    <collection href="PlannedOutages">
      <atom:title>PlannedOutages</atom:title>
    </collection>
    <collection href="Customers">
      <atom:title>Customers</atom:title>
    </collection>
    <collection href="HierarchyGroups">
      <atom:title>HierarchyGroups</atom:title>
    </collection>
    <collection href="RecentHistory">
      <atom:title>RecentHistory</atom:title>
    </collection>
    <collection href="RestorationSteps">
      <atom:title>RestorationSteps</atom:title>
    </collection>
    <collection href="ChronologyEntries">
      <atom:title>ChronologyEntries</atom:title>
    </collection>
    <collection href="Polygons">
      <atom:title>Polygons</atom:title>
    </collection>
    <collection href="AreaNotes">
      <atom:title>AreaNotes</atom:title>
    </collection>
  </workspace>
</service>
```

```

    </collection>
    <collection href="UnassociatedCalls">
      <atom:title>UnassociatedCalls</atom:title>
    </collection>
    <collection href="LastUpdate">
      <atom:title>LastUpdate</atom:title>
    </collection>
  </workspace>
</service>

```

2.1. Metadata of Web Service

Navigating to the URI "http://localhost:8800/OMS/\$metadata" lists all of the properties for all of the collections in the system. Detailed descriptions of these properties are provided in [Supported Service Types](#).

Some of the properties of Incident are shown in the code example below, which is displayed in edmx, using the Conceptual Schema Definition Language (CSDL).

Web Service Schema

```

<edmx:Edmx xmlns:edmx="http://schemas.microsoft.com/ado/2007/06/edmx" Version="1.0">
  <edmx:DataServices xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata"
m:DataServiceVersion="3.0" m:MaxDataServiceVersion="3.0">
    <Schema xmlns="http://schemas.microsoft.com/ado/2009/11/edm" Namespace="OmsUtilities">
      <EntityType Name="Incident">
        <Key>
          <PropertyRef Name="Id"/>
        </Key>
        <Property Name="DevicesWithNewlyClosedPhases" Type="Collection(Edm.String)"
Nullable="false"/>
        <Property Name="HasDevicesWithNewlyClosedPhases" Type="Edm.Boolean" Nullable="false"/>
        <Property Name="DevicesWithNewlyOpenedPhases" Type="Collection(Edm.String)"
Nullable="false"/>
        <Property Name="HasDevicesWithNewlyOpenedPhases" Type="Edm.Boolean" Nullable="false"/>
        <Property Name="Name" Type="Edm.String"/>
        <Property Name="ProtectiveStation" Type="Edm.String"/>
        <Property Name="ProtectiveDeviceId" Type="Edm.String"/>
        <Property Name="ProtectiveDeviceName" Type="Edm.String"/>
        <Property Name="Cleared" Type="Edm.Boolean" Nullable="false"/>
      </EntityType>
    </Schema>
  </edmx:DataServices>
</edmx:Edmx>

```

Similarly, all of the properties for other collections, such as Crew, Employee, Customer, and HierarchyGroups, are provided with the command http://localhost:8800/OMS/\$metadata.

2.2. Format of Web Service

Navigating to the Incidents resource URI "http://localhost:8800/Oms/Incidents" lists all of the incidents and values of the properties in AtomPub format.

Navigating to the Incidents resource URI "http://localhost:8800/Oms/Incidents?\$format=json" lists all of the incidents and values of the properties in JSON format.

The examples below show a set of values for an incident in both JSON and AtomPub formats.

Web Service Query Return in JSON Format

```
{
  "odata.metadata": "http://localhost:8800/oms/$metadata#Incidents",
  "value": [
    {
      "odata.type": "OmsUtilities.OutageIncident",
      "Locked": false,
      "SeqNumber": 7,
      "Id": "0d3726c8-309f-4ff0-bdab-81a44e1c2218",
      "Name": "2210000007",
      "IncidentType": 3,
      "IncidentStatus": 0,
      "IncidentSymbol": "0",
      "IsPendingClear": false,
      "IsTypePredicted": false,
      "IsMomentary": false,
      "IsManuallyConfirmed": false,
      "IsOutage": true,
      "IsNonOutage": false,
      "IncidentPtX": 932798.8,
      "IncidentPtY": 767424.7,
      "X": 932798.8,
      "Y": 767424.7,
      "Cancelled": false,
      "Cleared": false,
      "ClearingOperator": "",
      "ClearTime": "0001-01-01T00:00:00",
      "StartTime": "2018-08-09T04:47:14.6718964-07:00",
      "EndTime": "0001-01-01T00:00:00",
      "StatusStartTime": "2018-08-09T04:47:14.6718964-07:00",
      "AcknowledgedTime": "0001-01-01T00:00:00",
      "FirstCallTime": "0001-01-01T00:00:00",
      "FirstCallCustomer": "555240143555240143555240143",
      "FirstCallCustomerAddress": "7694 SAND AVE, Thorne Rosie C",
      "ETR": "0001-01-01T00:00:00",
      "ETRSource": "",
      "CalculatedETRSource": "",
      "ITR": "2018-08-09T07:00:00.7219495-07:00",
      "CustomerModifiedTime": "2018-08-09T04:47:14.9391849-07:00",
      "CurrentETR": "2018-08-09T07:00:00.7219495-07:00",
      "HasETR": false,
      "EtrSequenceNumber": 0,
      "IsEtrExpired": true,
      "EtrExpiringSoonTime": 15,
      "IsEtrExpiring": false,
      "IsEtrExpiringSoon": false,
      "Duration": 5813,
    }
  ]
}
```



```

"DurationInStatus": 5813,
"Station": "BAKER",
"FeederName": "F5461",
"AreaNoteId": "BAKER F5461",
"HasNonEvaluatorAssigned": true,
"WorkOrderId": "",
"ReceivedCompletionFromWfm": false,
"CrewId": "",
"JustCrewId": "",
"CrewName": "",
"CrewRadioNumber": "",
"EvaluatorId": "",
"EvaluatorName": "",
"EvaluatorRadioNumber": "",
"AssignedPtX": "",
"AssignedPtY": "",
"EvaluateCrewName": "",
"IsEvaluatorAssigned": false,
"CrewsAssignedCount": 0,
"CrewType": "",
"CaseNote": 301,
"ManualCaseNoteOverride": false,
"AreaNoteCode": 0,
"AreaNoteEtr": "0001-01-01T00:00:00",
"DispatcherComments": "",
"CrewComments": "",
"Remarks": "",
"CreatedAfterTheFact": false,
"CaseNoteRemarks": "",
"LineCrewNeeded": false,
"TreeCrewNeeded": false,
"FirstResponderNeeded": false,
"EventOfInterest": false,
"Reason": 0,
"NoTroubleFound": false,
"CancelOrder": false,
"SuppressNotification": false,
"SuppressNotificationReason": 0,
"SystemModeAreaNote": 0,
"NoOfPolesDown": 0,
"NoServicesDown": 0,
"NoOfTransformers": 0,
"ConductorSpan": 0,
"FollowupRequired": false,
"Inaccessible": false,
"PhaseOut": 0,
"OutageDeviceType": 4,
"ActionTaken": 0,
"PowerQualityType": 0,
"NonoutageType": 0,
"Weather": 0,
"SystemAffected": 0,
"FailureMode": 0,

```

```

"Manufacturer": 0,
"Animals": 0,
"Plants": 0,
"OutageType": 0,
"Material": 0,
"FailedEquipment": 0,
"SystemType": 1,
"OrgLevel1": "WSTC",
"OrgLevel2": "LEVITT",
"OrgLevel3": "GORDON",
"OrgLevel4": "GE",
"OrgLevel5": "",
"OrgLevel6": "",
"OrgLevel7": "",
"OrgLevel8": "",
"OrgLevel9": "",
"AssessmentLocationsCount": 0,
"ProtectiveStation": "BAKER",
"ProtectiveDeviceId": "60283538",
"ProtectiveDeviceName": "53007",
"CallType": 0,
"Condition_1": 0,
"Condition_2": 0,
"CallDescription": 0,
"CallCount": 0,
"AMICount": 0,
"AMIOutCount": 0,
"AMIIInCount": 0,
"IsPredictedStable": false,
"IsCallBack": false,
"CanConvertIncident": false,
"IncidentClearAttemptFailed": false,
"PointOfFailure": "",
"OpenDevicePtX": 932698.8,
"OpenDevicePtY": 767324.7,
"LockingUserName": "",
"ViewPortName": "",
"WorkStation": "",
"OpenDeviceStation": "BAKER",
"OpenDeviceId": "60283538",
"OpenDeviceName": "53007",
"SwitchingOrderId": "",
"SwitchingOrderName": "",
"OpenDeviceType": 4,
"OpenSwitchType": 1,
"CriticalCustomerCount": 0,
"ExcludeFromStatistics": false,
"StationNameDeviceId": "BAKER_53007_60283538",
"Cmi": 75569,
"PriorityCallsCount": 0,
"AssistanceNeeded": false,
"LastAmiInTimeReported": "0001-01-01T00:00:00",
"OpenDeviceIsScadaDevice": false,

```

```

    "CallCountPhaseA": 0,
    "CallCountPhaseB": 0,
    "CallCountPhaseC": 0,
    "CallCountPhaseAB": 0,
    "CallCountPhaseBC": 0,
    "CallCountPhaseAC": 0,
    "CallCountPhaseABC": 0,
    "MidpointSwitching": false,
    "OpenDeviceVoltage": 13.2790561913614,
    "CustomerCount": 13,
    "AmiCallCount": 0,
    "IsScada": false,
    "MeasName": "",
    "FromCallback": false,
    "OriginalDeviceName": "53007",
    "OriginalDeviceId": "60283538",
    "OriginalDeviceType": 4,
    "OriginalSwitchType": 1,
    "Division": "LEVITT",
    "FunctionalPermission": null,
    "OpCenter": "WSTC",
    "ServiceCenter": "WSTC",
    "PriorityCode": 0,
    "RankingManualOverride": 0,
    "RankingOverridden": false,
    "Ranking": 17,
    "MergedNonOutageCalls": [],
    "AffectedCustomersCount": 13,
    "CallAggregation1": 0,
    "CallAggregation2": 0,
    "CallAggregation3": 0,
    "CallAggregation4": 0,
    "CallAggregation5": 0,
    "CallAggregation6": 0,
    "CallAggregation7": 0,
    "CallAggregation8": 0,
    "CallAggregation9": 0,
    "CallAggregation10": 0,
    "CurrentCustomer": 13,
    "CurrentCriticalCustomer": 0,
    "HasHazardousCustomer": false,
    "HasRestorationSteps": true,
    "Created": "2018-08-09T04:47:14.6718964-07:00",
    "Modified": "2018-08-09T04:47:14.9381842-07:00"
  },
  {
    "odata.type": "OmsUtilities.OutageIncident",
    "Locked": false,
    "SeqNumber": 3,
    "Id": "1051930b-b2b1-4d27-a3b3-99b21e8e0981",
    ...

```

Web Service Query Return in AtomPub Format

```

<?xml version="1.0" encoding="utf-8"?>
<feed xml:base="http://localhost:8800/oms/"
      xmlns="http://www.w3.org/2005/Atom"
      xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices"
      xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata"
      xmlns:georss="http://www.georss.org/georss"
      xmlns:gml="http://www.opengis.net/gml">
  <id>http://localhost:8800/Oms/Incidents</id>
  <title type="text">Incidents</title>
  <updated>2018-08-13T12:41:31Z</updated>
  <link rel="self" title="Incidents" href="Incidents" />
  <entry>
    <id>http://localhost:8800/oms/Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')</id>
    <category term="OmsUtilities.OutageIncident"
      scheme="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme" />
    <link rel="edit" title="Incident" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident" />
    <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/ChronologyLog"
      type="application/atom+xml;type=feed" title="ChronologyLog" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident/ChronologyLog" />
    <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/RestorationSteps"
      type="application/atom+xml;type=feed" title="RestorationSteps" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident/RestorationSteps" />
    <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/Customers"
      type="application/atom+xml;type=feed" title="Customers" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident/Customers" />
    <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/UnassociatedCalls"
      type="application/atom+xml;type=feed" title="UnassociatedCalls" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident/UnassociatedCalls" />
    <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/SwitchingSteps"
      type="application/atom+xml;type=feed" title="SwitchingSteps" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident/SwitchingSteps" />
    <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/CustomFields"
      type="application/atom+xml;type=feed" title="CustomFields" href="Incidents('0d3726c8-309f-4ff0-bdab-81a44e1c2218')/OmsUtilities.OutageIncident/CustomFields" />
    <title />
    <updated>2018-08-13T12:41:31Z</updated>
    <author>
      <name />
    </author>
    <content type="application/xml">
      <m:properties>
        <d:Locked m:type="Edm.Boolean">false</d:Locked>
        <d:SeqNumber m:type="Edm.Int32">7</d:SeqNumber>
        <d:Id>0d3726c8-309f-4ff0-bdab-81a44e1c2218</d:Id>
        <d:Name>2210000007</d:Name>
        <d:IncidentType m:type="Edm.Int32">3</d:IncidentType>
        <d:IncidentStatus m:type="Edm.Int32">0</d:IncidentStatus>
        <d:IncidentSymbol>0</d:IncidentSymbol>
        <d:IsPendingClear m:type="Edm.Boolean">false</d:IsPendingClear>
        <d:IsTypePredicted m:type="Edm.Boolean">false</d:IsTypePredicted>
        <d:IsMomentary m:type="Edm.Boolean">false</d:IsMomentary>
        <d:IsManuallyConfirmed m:type="Edm.Boolean">false</d:IsManuallyConfirmed>
      </m:properties>
    </content>
  </entry>
</feed>

```

```

    <d:IsOutage m:type="Edm.Boolean">true</d:IsOutage>
    <d:IsNonOutage m:type="Edm.Boolean">false</d:IsNonOutage>
    <d:IncidentPtX m:type="Edm.Single">932798.8</d:IncidentPtX>
    <d:IncidentPtY m:type="Edm.Single">767424.7</d:IncidentPtY>
    <d:X m:type="Edm.Single">932798.8</d:X>
    <d:Y m:type="Edm.Single">767424.7</d:Y>
    <d:Cancelled m:type="Edm.Boolean">false</d:Cancelled>
    <d:Cleared m:type="Edm.Boolean">false</d:Cleared>
    <d:ClearingOperator></d:ClearingOperator>
    <d:ClearTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:ClearTime>
    <d:StartTime m:type="Edm.DateTime">2018-08-09T04:47:14.6718964-07:00</d:StartTime>
    <d:EndTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:EndTime>
    <d>StatusStartTime m:type="Edm.DateTime">2018-08-09T04:47:14.6718964-
07:00</d>StatusStartTime>
    <d:AcknowledgedTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:AcknowledgedTime>
    <d:FirstCallTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:FirstCallTime>
    <d:FirstCallCustomer>555240143555240143555240143</d:FirstCallCustomer>
    <d:FirstCallCustomerAddress>7694 SAND AVE, Thorne Rosie C</d:FirstCallCustomerAddress>
    <d:ETR m:type="Edm.DateTime">0001-01-01T00:00:00</d:ETR>
    <d:ETRSource></d:ETRSource>
    <d:CalculatedETRSource></d:CalculatedETRSource>
    <d:ITR m:type="Edm.DateTime">2018-08-09T07:00:00.7219495-07:00</d:ITR>
    <d:CustomerModifiedTime m:type="Edm.DateTime">2018-08-09T04:47:14.9391849-
07:00</d:CustomerModifiedTime>
    <d:CurrentETR m:type="Edm.DateTime">2018-08-09T07:00:00.7219495-07:00</d:CurrentETR>
    <d:HasETR m:type="Edm.Boolean">false</d:HasETR>
    <d:EtrSequenceNumber m:type="Edm.Int32">0</d:EtrSequenceNumber>
    <d:IsEtrExpired m:type="Edm.Boolean">true</d:IsEtrExpired>
    <d:EtrExpiringSoonTime m:type="Edm.Int32">15</d:EtrExpiringSoonTime>
    <d:IsEtrExpiring m:type="Edm.Boolean">false</d:IsEtrExpiring>
    <d:IsEtrExpiringSoon m:type="Edm.Boolean">false</d:IsEtrExpiringSoon>
    <d:Duration m:type="Edm.Int32">5814</d:Duration>
    <d:DurationInStatus m:type="Edm.Int32">5814</d:DurationInStatus>
    <d:Station>BAKER</d:Station>
    <d:FeederName>F5461</d:FeederName>
    <d:AreaNoteId>BAKER F5461</d:AreaNoteId>
    <d:HasNonEvaluatorAssigned m:type="Edm.Boolean">true</d:HasNonEvaluatorAssigned>
    <d:WorkOrderId></d:WorkOrderId>
    <d:ReceivedCompletionFromWfm m:type="Edm.Boolean">false</d:ReceivedCompletionFromWfm>
    <d:CrewId></d:CrewId>
    <d:JustCrewId></d:JustCrewId>
    <d:CrewName></d:CrewName>
    <d:CrewRadioNumber></d:CrewRadioNumber>
    <d:EvaluatorId></d:EvaluatorId>
    <d:EvaluatorName></d:EvaluatorName>
    <d:EvaluatorRadioNumber></d:EvaluatorRadioNumber>
    <d:AssignedPtX></d:AssignedPtX>
    <d:AssignedPtY></d:AssignedPtY>
    <d:EvaluateCrewName></d:EvaluateCrewName>
    <d:IsEvaluatorAssigned m:type="Edm.Boolean">false</d:IsEvaluatorAssigned>
    <d:CrewsAssignedCount m:type="Edm.Int32">0</d:CrewsAssignedCount>
    <d:CrewType></d:CrewType>
    <d:CaseNote m:type="Edm.Int32">301</d:CaseNote>

```

```

<d:ManualCaseNoteOverride m:type="Edm.Boolean">false</d:ManualCaseNoteOverride>
<d:AreaNoteCode m:type="Edm.Int32">0</d:AreaNoteCode>
<d:AreaNoteEtr m:type="Edm.DateTime">0001-01-01T00:00:00</d:AreaNoteEtr>
<d:DispatcherComments></d:DispatcherComments>
<d:CrewComments></d:CrewComments>
<d:Remarks></d:Remarks>
<d:CreatedAfterTheFact m:type="Edm.Boolean">false</d:CreatedAfterTheFact>
<d:CaseNoteRemarks></d:CaseNoteRemarks>
<d:LineCrewNeeded m:type="Edm.Boolean">false</d:LineCrewNeeded>
<d:TreeCrewNeeded m:type="Edm.Boolean">false</d:TreeCrewNeeded>
<d:FirstResponderNeeded m:type="Edm.Boolean">false</d:FirstResponderNeeded>
<d:EventOfInterest m:type="Edm.Boolean">false</d:EventOfInterest>
<d:Reason m:type="Edm.Int32">0</d:Reason>
<d:NoTroubleFound m:type="Edm.Boolean">false</d:NoTroubleFound>
<d:CancelOrder m:type="Edm.Boolean">false</d:CancelOrder>
<d:SuppressNotification m:type="Edm.Boolean">false</d:SuppressNotification>
<d:SuppressNotificationReason m:type="Edm.Int32">0</d:SuppressNotificationReason>
<d:SystemModeAreaNote m:type="Edm.Int32">0</d:SystemModeAreaNote>
<d:NoOfPolesDown m:type="Edm.Int32">0</d:NoOfPolesDown>
<d:NoServicesDown m:type="Edm.Int32">0</d:NoServicesDown>
<d:NoOfTransformers m:type="Edm.Int32">0</d:NoOfTransformers>
<d:ConductorSpan m:type="Edm.Int32">0</d:ConductorSpan>
<d:FollowupRequired m:type="Edm.Boolean">false</d:FollowupRequired>
<d:Inaccessible m:type="Edm.Boolean">false</d:Inaccessible>
<d:PhaseOut m:type="Edm.Int32">0</d:PhaseOut>
<d:OutageDeviceType m:type="Edm.Int32">4</d:OutageDeviceType>
<d>ActionTaken m:type="Edm.Int32">0</d>ActionTaken>
<d:PowerQualityType m:type="Edm.Int32">0</d:PowerQualityType>
<d:NonoutageType m:type="Edm.Int32">0</d:NonoutageType>
<d:Weather m:type="Edm.Int32">0</d:Weather>
<d:SystemAffected m:type="Edm.Int32">0</d:SystemAffected>
<d:FailureMode m:type="Edm.Int32">0</d:FailureMode>
<d:Manufacturer m:type="Edm.Int32">0</d:Manufacturer>
<d:Animals m:type="Edm.Int32">0</d:Animals>
<d:Plants m:type="Edm.Int32">0</d:Plants>
<d:OutageType m:type="Edm.Int32">0</d:OutageType>
<d:Material m:type="Edm.Int32">0</d:Material>
<d:FailedEquipment m:type="Edm.Int32">0</d:FailedEquipment>
<d:SystemType m:type="Edm.Int32">1</d:SystemType>
<d:OrgLevel1>WSTC</d:OrgLevel1>
<d:OrgLevel2>LEVITT</d:OrgLevel2>
<d:OrgLevel3>GORDON</d:OrgLevel3>
<d:OrgLevel4>GE</d:OrgLevel4>
<d:OrgLevel5></d:OrgLevel5>
<d:OrgLevel6></d:OrgLevel6>
<d:OrgLevel7></d:OrgLevel7>
<d:OrgLevel8></d:OrgLevel8>
<d:OrgLevel9></d:OrgLevel9>
<d:AssessmentLocationsCount m:type="Edm.Int32">0</d:AssessmentLocationsCount>
<d:ProtectiveStation>BAKER</d:ProtectiveStation>
<d:ProtectiveDeviceId>60283538</d:ProtectiveDeviceId>
<d:ProtectiveDeviceName>53007</d:ProtectiveDeviceName>
<d:CallType m:type="Edm.Int32">0</d:CallType>

```

```

    <d:Condition_1 m:type="Edm.Int32">0</d:Condition_1>
    <d:Condition_2 m:type="Edm.Int32">0</d:Condition_2>
    <d:CallDescription m:type="Edm.Int32">0</d:CallDescription>
    <d:CallCount m:type="Edm.Int32">0</d:CallCount>
    <d:AMICount m:type="Edm.Int32">0</d:AMICount>
    <d:AMIOutCount m:type="Edm.Int32">0</d:AMIOutCount>
    <d:AMIIInCount m:type="Edm.Int32">0</d:AMIIInCount>
    <d:IsPredictedStable m:type="Edm.Boolean">>false</d:IsPredictedStable>
    <d:IsCallBack m:type="Edm.Boolean">>false</d:IsCallBack>
    <d:CanConvertIncident m:type="Edm.Boolean">>false</d:CanConvertIncident>
    <d:IncidentClearAttemptFailed
m:type="Edm.Boolean">>false</d:IncidentClearAttemptFailed>
    <d:PointOfFailure></d:PointOfFailure>
    <d:OpenDevicePtX m:type="Edm.Single">932698.8</d:OpenDevicePtX>
    <d:OpenDevicePtY m:type="Edm.Single">767324.7</d:OpenDevicePtY>
    <d:LockingUserName></d:LockingUserName>
    <d:ViewPortName></d:ViewPortName>
    <d:WorkStation></d:WorkStation>
    <d:OpenDeviceStation>BAKER</d:OpenDeviceStation>
    <d:OpenDeviceId>60283538</d:OpenDeviceId>
    <d:OpenDeviceName>53007</d:OpenDeviceName>
    <d:SwitchingOrderId></d:SwitchingOrderId>
    <d:SwitchingOrderName></d:SwitchingOrderName>
    <d:OpenDeviceType m:type="Edm.Int32">4</d:OpenDeviceType>
    <d:OpenSwitchType m:type="Edm.Int32">1</d:OpenSwitchType>
    <d:CriticalCustomerCount m:type="Edm.Int32">0</d:CriticalCustomerCount>
    <d:ExcludeFromStatistics m:type="Edm.Boolean">>false</d:ExcludeFromStatistics>
    <d:StationNameDeviceId>BAKER_53007_60283538</d:StationNameDeviceId>
    <d:Cmi m:type="Edm.Int32">75582</d:Cmi>
    <d:PriorityCallsCount m:type="Edm.Int32">0</d:PriorityCallsCount>
    <d:AssistanceNeeded m:type="Edm.Boolean">>false</d:AssistanceNeeded>
    <d>LastAmiInTimeReported m:type="Edm.DateTime">0001-01-
01T00:00:00</d>LastAmiInTimeReported>
    <d:OpenDeviceIsScadaDevice m:type="Edm.Boolean">>false</d:OpenDeviceIsScadaDevice>
    <d:CallCountPhaseA
...

```

2.3. Filtering Web Services

Results can be filtered using the "filter" keyword. For example, the URI:

[http://localhost:8800/Oms/Incidents?\\$filter=Name eq '0500000076'](http://localhost:8800/Oms/Incidents?$filter=Name eq '0500000076')

returns only the incident with the Name "0500000076".

Web Service Query Using "filter" Keyword

```

<?xml version="1.0" encoding="utf-8"?>
<feed xml:base="http://localhost:8800/oms/"
      xmlns="http://www.w3.org/2005/Atom"
      xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices"

```

```

xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata">
<id>http://localhost:8800/Oms/Incidents</id>
<title type="text">Incidents</title>
<updated>2017-02-19T22:13:48Z</updated>
<link rel="self" title="Incidents" href="Incidents" />
<entry>
  <id>http://localhost:8800/oms/Incidents('16b12494-0cb6-44a0-b3e7-f72b1ff42fdf')</id>
  <category term="OmsUtilities.Incident"
scheme="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme" />
  <link rel="edit" title="Incident" href="Incidents('16b12494-0cb6-44a0-b3e7-f72b1ff42fdf')"/>
  <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/RestorationSteps"
type="application/atom+xml;type=feed" title="RestorationSteps" href="Incidents('16b12494-0cb6-44a0-
b3e7-f72b1ff42fdf')/RestorationSteps" />
  <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/ChronologyLog"
type="application/atom+xml;type=feed" title="ChronologyLog" href="Incidents('16b12494-0cb6-44a0-b3e7-
f72b1ff42fdf')/ChronologyLog" />
  <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/Customers"
type="application/atom+xml;type=feed" title="Customers" href="Incidents('16b12494-0cb6-44a0-b3e7-
f72b1ff42fdf')/Customers" />
  <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/UnassociatedCalls"
type="application/atom+xml;type=feed" title="UnassociatedCalls" href="Incidents('16b12494-0cb6-44a0-
b3e7-f72b1ff42fdf')/UnassociatedCalls" />
  <link rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/SwitchingSteps"
type="application/atom+xml;type=feed" title="SwitchingSteps" href="Incidents('16b12494-0cb6-44a0-b3e7-
f72b1ff42fdf')/SwitchingSteps" />
  <title />
  <updated>2017-02-19T22:13:48Z</updated>
  <author>
    <name />
  </author>
  <content type="application/xml">
    <m:properties>
      <d:Name>0500000076</d:Name>
      <d:AssessmentLocationsCount m:type="Edm.Integer">0</d:AssessmentLocationsCount>
      <d:ProtectiveStation>RAVEN</d:ProtectiveStation>
      <d:ProtectiveDeviceId>17882948</d:ProtectiveDeviceId>
      <d:ProtectiveDeviceName>6-7313-7361-0-1</d:ProtectiveDeviceName>
      <d:Cleared m:type="Edm.Boolean">>false</d:Cleared>
      <d:StartTime m:type="Edm.DateTime">2017-02-19T11:06:04.1296722-08:00</d:StartTime>
      <d:AcknowledgedTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:AcknowledgedTime>
      <d:FirstCallTime m:type="Edm.DateTime">2017-02-19T11:06:04.1296722-
08:00</d:FirstCallTime>
      <d:CallCount m:type="Edm.Integer">2</d:CallCount>
      <d:AMICount m:type="Edm.Integer">0</d:AMICount>
      <d:AMIOutCount m:type="Edm.Integer">0</d:AMIOutCount>
      <d:AMIIInCount m:type="Edm.Integer">0</d:AMIIInCount>
      <d:IsPredictedStable m:type="Edm.Boolean">>true</d:IsPredictedStable>
      <d:IsCallBack m:type="Edm.Boolean">>false</d:IsCallBack>
      <d:Station>RAVEN</d:Station>
      <d:FeederName>F8865</d:FeederName>
      <d:Cancelled m:type="Edm.Boolean">>false</d:Cancelled>
      <d:ReceivedCompletionFromWfm m:type="Edm.Boolean">>false</d:ReceivedCompletionFromWfm>
      <d:Id>16b12494-0cb6-44a0-b3e7-f72b1ff42fdf</d:Id>

```



```

<d:PointOfFailure></d:PointOfFailure>
<d:OpenDevicePtX m:type="Edm.Single">924820.1</d:OpenDevicePtX>
<d:OpenDevicePtY m:type="Edm.Single">818279.8</d:OpenDevicePtY>
<d:OpenDeviceStation>RAVEN</d:OpenDeviceStation>
<d:OpenDeviceId>17882948</d:OpenDeviceId>
<d:OpenDeviceName>6-7313-7361-0-1</d:OpenDeviceName>
<d:OpenDeviceType m:type="Edm.Integer">16</d:OpenDeviceType>
<d:OpenSwitchType m:type="Edm.Integer">-1</d:OpenSwitchType>
<d:CriticalCustomerCount m:type="Edm.Integer">0</d:CriticalCustomerCount>
<d:ExcludeFromStatistics m:type="Edm.Boolean">>false</d:ExcludeFromStatistics>
<d:StationNameDeviceId>RAVEN_6-7313-7361-0-1_17882948</d:StationNameDeviceId>
<d:Cmi m:type="Edm.Integer">1683</d:Cmi>
<d:WorkOrderId></d:WorkOrderId>
<d:PriorityCallsCount m:type="Edm.Integer">0</d:PriorityCallsCount>
<d:IsTypePredicted m:type="Edm.Boolean">>true</d:IsTypePredicted>
<d:IsMomentary m:type="Edm.Boolean">>false</d:IsMomentary>
<d:IsManuallyConfirmed m:type="Edm.Boolean">>false</d:IsManuallyConfirmed>
<d:IncidentStatus m:type="Edm.Integer">0</d:IncidentStatus>
<d:ClearTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:ClearTime>
<d:ClearingOperator></d:ClearingOperator>
<d:AssistanceNeeded m:type="Edm.Boolean">>false</d:AssistanceNeeded>
<d:EndTime m:type="Edm.DateTime">0001-01-01T00:00:00</d:EndTime>
<d>LastAmiInTimeReported m:type="Edm.DateTime">0001-01-
01T00:00:00</d>LastAmiInTimeReported>
<d:OpenDeviceIsScadaDevice m:type="Edm.Boolean">>false</d:OpenDeviceIsScadaDevice>
<d:CustomerCount m:type="Edm.Integer">9</d:CustomerCount>
<d:AmiCallCount m:type="Edm.Integer">0</d:AmiCallCount>
<d:IsEtrExpired m:type="Edm.Boolean">>true

```

2.4. Selecting a Field for a Query

If only a few properties of an object need to be retrieved, the "select" operator can be used to reduce the amount of data returned. For example, the following URI returns a list of crews with only the Division and Name properties having values:

```
http://localhost:8800/OMS/Crews?$select=Division,Name
```

2.5. Selecting an Item for a Query

If only data for particular items needs to be retrieved, the URI can be constructed by placing the key value for the item in parentheses. For example, the following URI returns data for the 3LIONSSC service center only:

```
http://localhost:8800/oms/HierarchyGroups('ORGLLEVEL1%203LIONSSC')
```

2.6. Navigation Properties

Navigation properties are used to get child objects from an object that contains links. For instance, Incident provides a navigation link to a collection, such as RestorationSteps or Customers, that are associated with that incident. The URI can be constructed by placing the key value for the object in parentheses and adding the navigation property, similar to:

```
http://localhost:8800/Oms/Incidents('IncidentId')/Customers
```

The available navigation properties are also returned in the query response with the "link" element.

```
<?xml version="1.0" encoding="utf-8"?>
<feed xml:base="http://localhost:8800/Oms/"
      xmlns="http://www.w3.org/2005/Atom"
      xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices"
      xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata">
  <id>http://localhost:8800/Oms/Incidents('2180000002')/Customers</id>
  <title type="text">Customers</title>
  <updated>2016-08-05T20:25:23Z</updated>
  <link rel="self" title="Customers" href="Customers" />
  <author>
    <name />
  </author>
</feed>
```

The query response when using navigation properties is a collection of the child object; the parent object is not returned. For example, to show a collection of employee data for the FI1564 crew in JSON format, use the following URI:

```
http://localhost:8800/Oms/Crews('FI1564')/Employees?$format=json
```

The response is as follows:

```
{
  "odata.metadata": "http://localhost:8800/oms/$metadata#Employees",
  "value": [
    {
      "Id": "FI1564",
      "Name": "Ron X Stanton",
      "NameId": "Ron X Stanton ( FI1564 )",
      "FirstName": "Ron X",
      "MiddleName": "",
      "LastName": "Stanton",
      ...
      "IsForeign": false,
      "DefaultLeadName": "",
      "CurrentLeadName": "",
      "UserId": ""
    }
  ]
}
```

```
}
```

2.7. Expanding a Query

The "expand" operator is similar to using navigation properties; however, it is possible to retrieve both the collection of child objects and the parent object at the same time.

To return an incident along with the collection of customers affected by the incident, the following URI results in a query response where the returned Incident object has a Customers field that is a collection of Customer objects:

```
http://localhost:8800/Oms/Incidents('IncidentId')?$expand=Customers
```

2.8. More Information

For an explanation of the protocol used and examples of how the data can be filtered based on different criteria, refer to <http://www.odata.org/>.

The following application is helpful for building and testing queries: <http://www.silverlight.net/content/samples/odataexplorer/>.

3. Web Service Client

3.1. Client Overview

In general, the WCF Data Service client must be able to perform HTTP GET requests on the server. Depending on the query, a data service may return an XML or JSON response. The client must be aware of the type of data being requested and be able to read the response type.

3.2. Dashboard

You can create your own dashboards for OMS data. The data service has been tested by creating sample dashboards using both GWT (Google Web Toolkit) and ASP .NET MVC. While there are many tools to build these applications, your choice provides a chance to use multiple programming languages (Java, JavaScript, and C#) to access the data service.

4. Supported Service Types

The following sections describe the types that are exposed at the root level.

4.1. Incidents

<http://localhost:8800/Oms/Incidents>

Table 1. Incidents

Field Name	Type	Nullable	Description
AOR	String	false	Area of responsibility (AOR) this incident belongs to.
ServiceCenter	String	true	ID of the service center the incident belongs to.
ServiceCenterName	String	true	Name of the service center the incident belongs to.
OrgLevel1Id	String	true	Service center the incident belongs to.
OrgLevel2Id	String	true	The first (lowest) hierarchy group above service center. By default, City.
OrgLevel3Id	String	true	The second hierarchy group above service center. By default, County.
OrgLevel4Id	String	true	The third hierarchy group above service center. By default, State.
OrgLevel5Id	String	true	The fourth hierarchy group above service center. By default, System.
OrgLevel6Id	String	true	The fifth hierarchy group above service center. Not defined by default.
OrgLevel7Id	String	true	The sixth hierarchy group above service center. Not defined by default.
OrgLevel8Id	String	true	The seventh hierarchy group above service center. Not defined by default.
OrgLevel9Id	String	true	The eighth hierarchy group above service center. Not defined by default.
OrgLevel1Name	String	true	Service center the incident belongs to.
OrgLevel2Name	String	true	The first (lowest) hierarchy group above service center. By default, City.
OrgLevel3Name	String	true	The second hierarchy group above service center. By default, County.
OrgLevel4Name	String	true	The third hierarchy group above service center. By default, State.
OrgLevel5Name	String	true	The fourth hierarchy group above service center. By default, System.
OrgLevel6Name	String	true	The fifth hierarchy group above service center. Not defined by default.

Field Name	Type	Nullable	Description
OrgLevel7Name	String	true	The sixth hierarchy group above service center. Not defined by default.
OrgLevel8Name	String	true	The seventh hierarchy group above service center. Not defined by default.
OrgLevel9Name	String	true	The eighth hierarchy group above Service center. Not defined by default.
RegionLevels	Collection	false	Information for incident regions (hierarchy groups), including region type, name, and ID.
SeqNumber	Integer	false	Sequential number of the incident in the current year.
SystemModeAreaNote	Integer	false	Numeric value of the System Mode of the incident, which could be one of the following: Normal = 0 Storm = 1 Major Storm = 2
Name	String	true	Name of the incident, which is a nine-digit value. The first three digits indicate the day of the year and the next seven digits represent the counter of the incident.
IncidentPtX	Single	false	X-coordinate of the incident location.
IncidentPtY	Single	false	Y-coordinate of the incident location.
DeviceX	Single	false	X-coordinate of the device of record.
DeviceY	Single	false	Y-coordinate of the device of record.
CustomerCount	Integer	false	Number of customers.
ProtectiveStation	String	true	Station name of the protective device.
ProtectiveDeviceId	String	true	ID of the protective device that tripped.
ProtectiveDeviceName	String	true	Name of the protective device that tripped.
Cleared	Boolean	false	Has the incident been cleared?
StartTime	DateTime	false	Start time of the incident.
FirstCallTime	DateTime	false	Timestamp of the first customer call.
FirstCallCustomer	String	true	The customer key of the customer who called first. For incidents without customer calls, this is the customer key of the first customer in the list of affected customers.
FirstCallCustomerAddress	String	true	The address and name of the customer who called first. For incidents without customer calls, this is the address and name of the first customer in the list of affected customers.
FirstCallComment	String	true	The comment from the customer who called first.
FirstCrewTime	DateTime	false	Date and time of the first crew assignment to the incident.

Field Name	Type	Nullable	Description
CallType	Integer	false	Type of the call that originated the incident. The numeric value is specified according to the CallType coded translation item. The default values are: None = 0 Lights Out = 1 Area Out = 2 Street Light = 3 ODL Light = 4 Dim/Bright = 5 Part Power = 6 Dim = 7 Blinking Lights = 8 Miscellaneous = 9 Still Out = 10
Condition_1	Integer	false	Numerical value of the first condition specified in the customer call, as configured by the analyst.
Condition_2	Integer	false	Numerical value of the second condition specified in the customer call, as configured by the analyst.
CallDescription	Integer	false	Numerical value of the description specified in the customer call, as configured by the analyst.
CallCount	Integer	false	Number of calls for this incident.
AMICount	Integer	false	Number of AMI calls for the incident.
IsCallBack	Boolean	false	Indicates if callback is required.
Station	String	true	Name of the nominal substation of the device of record.
FeederId	String	true	Id of the nominal feeder of the device of record.
FeederName	String	true	Name of the nominal feeder of the device of record.
OperatedStation	String	true	Name of the energizing substation of the device of record.
OperatedFeederName	String	true	Name of the energizing feeder of the device of record.
Cancelled	Boolean	false	Has the incident been cancelled?
Id	Id	false	ID of the incident.
OriginalDevicePtX	Float	false	X-coordinate of the location of the initially affected device.
OriginalDevicePtY	Float	false	Y-coordinate of the location of the initially affected device.
OpenDevicePtX	Single	false	X-coordinate of the open device's location.
OpenDevicePtY	Single	false	Y-coordinate of the open device's location.
OpenDeviceStation	String	true	Station name where the open device belongs to.
OpenDeviceId	String	true	ID of the last opened device from an incident.
OpenDeviceName	String	true	Name of the last opened device from an incident.

Field Name	Type	Nullable	Description
OpenDeviceType	Integer	false	Type of the device that is currently open.
OpenSwitchType	Integer	false	Type of the switching device that is currently open.
IsOverhead	Boolean	false	Specifies whether the device of record is overhead.
CriticalCustomerCount	Integer	false	Number of critical customers interrupted.
ExcludeFromStatistics	Boolean	false	Exclude from the calculation of outage statistics.
StationNameDeviceId	String	true	Provides both station name and device ID together.
Cmi	Integer	false	CMI of the incidents.
PriorityCallsCount	Integer	false	Number of priority calls, as configured on the "Is a Priority Call" tab in the OMS Server Configuration Editor.
IsMomentary	Boolean	false	Indicates if the incident's type is Momentary.
IsManuallyConfirmed	Boolean	false	Indicates if the incident's type is Manually Confirmed.
IncidentStatus	Integer	false	Numeric value of the working status of the incident, which could be one of the following: Unassigned = 0 SentToWfm = 1 PendingAssigned = 2 Assigned = 3 EnRoute = 4 OnSite = 5 Cancelled = 6 Complete = 7 PendingClear = 8 Patrolled = 9 Evaluating = 10 Evaluated = 11
IncidentSubtype	Integer	false	Numeric value of the incident subtype. The value is based on the PlannedOutageType or ConfirmedIncidentSubtype coded translation item.
IncidentSymbol	Integer	false	Numeric value of the incident symbol, which could be one of the following: Unassigned = 0 SentToWfm = 1 PendingAssigned = 2 Assigned = 3 EnRoute = 4 OnSite = 5 Cancelled = 6 Complete = 7 PendingClear = 8 Patrolled = 9 Evaluating = 10 Evaluated = 11 NonOutageIncident = 12
ClearTime	DateTime	false	Time when the Incident was cleared.

Field Name	Type	Nullable	Description
ClearingOperator	Boolean	true	Indicates if the incident can be cleared by the operator.
AssistanceNeeded	Boolean	false	Crew is waiting for assistance for this incident.
EndTime	DateTime	false	Time when the incident ended.
LastAmiInTimeReported	DateTime	false	Last time there was an Ami in message received from the Ami system.
OpenDeviceIsScadaDevice	Boolean	false	The device of origin opened for the incident is a SCADA device.
CallCountPhaseA	Integer	false	The number of phase A customer calls.
CallCountPhaseB	Integer	false	The number of phase B customer calls.
CallCountPhaseC	Integer	false	The number of phase C customer calls.
CallCountPhaseAB	Integer	false	The number of phase AB customer calls.
CallCountPhaseBC	Integer	false	The number of phase BC customer calls.
CallCountPhaseAC	Integer	false	The number of phase AC customer calls.
CallCountPhaseABC	Integer	false	The number of phase ABC customer calls.
MidpointSwitching	Boolean	false	Specifies whether this is a midpoint switching.
OpenDeviceVoltage	Double	false	The voltage of the device of record.
OpenDeviceVoltageType	Boolean	false	The voltage type of the device of record, where true = LL (Line-to-Line) and false = LG (Line-to-Ground).
TotalCriticalCustomerCount	Integer	false	The number of critical customers of all types.
CriticalCustomerCount	Integer	false	The number of type 1 critical customers.
CriticalCustomerCount2	Integer	false	The number of type 2 critical customers.
CriticalCustomerCount3	Integer	false	The number of type 3 critical customers.
CriticalCustomerCount4	Integer	false	The number of type 4 critical customers.
CriticalCustomerCount5	Integer	false	The number of type 5 critical customers.
AmiCallCount	Integer	false	Number of Ami Out messages received.
EtrSequenceNumber	Integer	false	Shows the sequential number for ETR changes. The sequential number starts with 0 and increments by 1 each time the incident ETR changes (for example, when assigning a crew or creating an area setting).
IsEtrExpired	Boolean	false	Is ETR already expired on this incident?
EtrExpiringSoonTime	Integer	false	Time in minutes before the ETR expiration time for the incident to be considered as expiring soon.
IsEtrExpiring	Boolean	false	Is ETR expiring on this incident? It is a configurable parameter set by the analyst.
IsEtrExpiringSoon	Boolean	false	Indicates if the ETR for the incident is expiring soon.

Field Name	Type	Nullable	Description
CaseNote	Integer	false	Case note of the incident. Numeric value is one of the following: 201: No Case Note Assigned (default message) 301: Outage recorded – crew en route 302: Trouble located – crew working to restore power 303: Outage recorded – crew assigned not dispatched 304: Trouble located – crew dispatched 305: Large number of outages – crews not available 306: Vehicle accident 307: Underground cable damaged 308: A tree has fallen on lines 309: Lightning and thunderstorms 310: High winds caused numerous outages 311: Trouble fixed – power will be restored soon 312: Power has been restored 401: Planned interruption
CaseNote	String	true	Case note description.
ManualCaseNoteOverride	Boolean	false	Has a case note been manually overridden for the incident?
AreaNoteId	String	false	The name of the lowest management area that is related to the area setting.
AreaNoteCode	Integer	String	Area note code, according to the AreaNote coded translation item.
AreaNoteText	String	true	Area note text, according to the AreaNote coded translation item.
AreaNoteFreeFormRemarks	String	true	Area note custom text value specified in the Use Message field on the Update Area Settings display.
AreaNoteEtr	DateTime	false	End time of the area setting.
HasCrewWithBlockingStatus	Boolean	false	Specifies whether a crew with working status that blocks ETR override by an area setting is assigned to the incident.
StormName	String	true	The name of the storm that is associated with the incident's area setting.
DynAreaNoteWorkingId	String	true	The current area setting's ID concatenated with the NoteStamp value.
IsThirdParty	Boolean	false	Indicates whether the device of record belongs to a third-party company.
AreaNoteStartTime	DateTime	false	The start time of the incident's area setting.
AreaNoteITRDisable	Boolean	false	Indicates if setting incident ITR in the area setting's area is disabled.
IsScada	Boolean	false	Indicates if the device is a SCADA device.
FromCallback	Boolean	false	Is the incident the result of a callback?
FoundOpen	Boolean	false	Specifies whether the device of record was found open in field.

Field Name	Type	Nullable	Description
OriginalDeviceName	String	true	Name of the original device that caused the incident.
OriginalDeviceId	String	true	ID of the device that caused the incident.
OriginalDeviceType	Integer	false	Type of the device that caused the incident.
OriginalSwitchType	Integer	false	Switch type of the device that caused the incident.
X	Single	false	X-coordinate of the incident.
Y	Single	false	Y-coordinate of the incident.
Division	String	true	Division this incident belongs to.
WorkOrg	String	true	Name of the assigned work organization, if assigned.
WorkOrg2	String	true	Name of the second assigned work organization, if assigned.
WorkOrg3	String	true	Name of the third assigned work organization, if assigned.
FunctionalPermission	String	true	Functional permission that is required to modify the incident.
HasNonEvaluatorAssigned	Boolean	false	Flag for a non-evaluator crew assigned to the incident.
OpCenter	String	true	Name of the service center the incident belongs to.
OperatedServiceCenter	String	true	Name of the service center the device of record dynamically belongs to.
LocationDescription	String	true	Information about the device location (for example, pole number or street number).
State	String	true	The name of the associated state.
County	String	true	The name of the associated county.
Municipality	String	true	The name of the associated municipality.
IncidentType	Integer	false	Type of the incident. The value is based on the IncidentType coded translation item. This could be one of the following numeric values: SingleCustomer = 0 Momentary = 1 Predicted = 2 Confirmed = 3 Isolated = 4 ManuallyConfirmed = 5 Planned = 6 NonOutage = 7
IncidentType2	Integer	false	Type of the incident. The value is based on the IncidentType coded translation item. The type is applicable in the OMS V2 architecture.
OutageStatus	Integer	false	Status of the outage. The value is based on the OutageStatus coded translation item. The status is applicable in the OMS V2 architecture.
PriorityCode	Integer	false	Highest call priority type of the incident calls.

Field Name	Type	Nullable	Description
Ranking	Integer	false	The calculated value of the incident ranking.
RankingManualOverride	Integer	false	The manually entered value of the incident ranking.
RankingOverridden	Boolean	false	Specifies whether the manually entered incident ranking is used instead of the calculated incident ranking.
FRP	Integer	false	The feeder restoration priority (FRP). For breaker incidents, when the area setting's UseFRP flag is set to True, FRP is set to the value specified in the station model. For other incidents, FRP is set to "-1".
Duration	Integer	false	Duration of the Incident in hours.
AffectedCustomersCount	Integer	false	The total number of customers (both energized and de-energized) that are affected by the incident.
PreviousAffectedCustomersCount	Integer	false	The total number of customers (both energized and de-energized) that were affected by the incident at the time of the last periodic processing of incident data. This field is used to check whether the number of affected customers has changed.
CallAggregation1-10	Integer	false	Shows the number of calls satisfying each of the call aggregation rules.
NoOfPolesDown	Integer	false	Number of poles down for this incident.
NoServicesDown	Integer	false	Count of services down from damage assessment provided by crew.
NoOfTransformers	Integer	false	Number of transformers down.
NoOfCrossArms	Integer	false	Number of cross arms down.
NoOfTreeTrimSpans	Integer	false	Number of tree trim spans.
ConductorSpan	Integer	false	Number of spans of wire down.
FollowupRequired	Boolean	false	Is follow up work required after the incident is cleared?
Inaccessible	Boolean	false	Device affected does not have an easy access. ETR could be more for such an incident.
PhaseOut	Integer	false	One of the following numeric values: ABC = 0 AB = 1 BC = 2 AC = 3 A =4 B=5 C=6

Field Name	Type	Nullable	Description
OutageDeviceType	Integer	false	One of the following numeric values: None = 0 Cut = 1 Fuse = 2 Jumper = 3 LineSwitch = 4 Recloser = 5 Sectionalizer = 6 Breaker = 7 Transformer = 8 NetworkProtector = 9 ServPoint = 10 TransformerBank = 11 Structure = 12 Equipment = 13 Elbow = 14
ActionTaken	Integer	false	Action taken to fix the problem. One of the following numeric values: None = 0 Rehung/Replaced = 1 Replaced Pole = 2 Replaced Transformer = 3
Material	Integer	false	Material affected by the incident. One of the following numeric values: None = 0 Other = 1 Recloser = 2 Regulator = 3 Transformer = 4 Arrester = 5 BracketCrossarm = 6 Capacitor = 7 ConductorOH = 8 ConnectorClamp = 9 ElbowLB = 10 Insulator = 11 Pole = 12 PrimaryTerminator = 13 Splice = 14 SwitchCutout = 15 UgCable = 16
FailedEquipment	Integer	false	Equipment affected by the incident.

Field Name	Type	Nullable	Description
OutageType	Integer	false	Outage type of the incident. One of the following numeric values: None = 0 CustomerTrouble = 1 ExternalSource = 2 Primary = 3 Secondary = 4 Service = 5 Substation = 6 Transformer = 7 Transmission = 8
CurrentCustomer	Integer	false	Number of customers currently de-energized because of the incident.
DispatcherComments	String	true	Dispatcher comments for the incident.
CrewComments	String	true	Crew comments for the incident.
CaseNoteRemarks	String	true	Case note remarks for the incident.
LineCrewNeeded	Boolean	false	Additional crew is needed to fix the line issue.
TreeCrewNeeded	Boolean	false	Additional crew is needed to fix the tree issue.
FirstResponderNeeded	Boolean	false	First responder is needed for the incident.
EvaluatorNeeded	Boolean	false	Evaluator is needed for the incident.
EventOfInterest	Boolean	false	Shows if the incident is an event of interest.
Reason	Integer	false	Reason of the incident. Returns a Reason coded translation item key.
ReasonGroup	Integer	false	Reason group of the incident. Returns a ReasonGroup coded translation item key.
NoTroubleFound	Boolean	false	Crew did not find any trouble for the reported incident.
CancelOrder	Boolean	false	Crew cancelled the order.
SuppressNotification	Boolean	false	Indicates whether automatic notifications are disabled for the incident's area.
CrewId	String	true	Crew IDs for all crews assigned to the incident (except for evaluator crews).
CrewName	String	true	Crew names for all crews assigned to the incident (except for evaluator crews).
CrewRadioNumber	String	true	Radio numbers for all crews assigned to the incident (except for evaluator crews).
EvaluatorId	String	true	Crew IDs of evaluator crews assigned to the incident.
EvaluatorName	String	true	Crew names for all evaluator crews assigned to the incident.
EvaluatorRadioNumber	String	true	Radio numbers for all evaluator crews assigned to the incident.
AssignedPtX	String	true	X-coordinate of the assigned crew location.

Field Name	Type	Nullable	Description
AssignedPtY	String	true	Y-coordinate of the assigned crew location.
CrewType	String	true	Crew types for all crews assigned to the incident.
CrewTypeIds	String	true	Crew type IDs for crews that are assigned to the incident.
ETR	DateTime	false	Estimated restoration time of the incident.
ITR	DateTime	false	Initial estimation for the restoration date and time. This value is set according to the ETR matrix when an incident is originated. ITR never changes.
HasETR	Boolean	false	Specifies whether the estimated restoration time of the incident is set.
IsAreaNoteAppliable	Boolean	false	Specifies whether an area setting applies to the incident. (True for active area settings and False for expired area settings.)
ETRSource	String	true	The name of the user or system that initiated the last ETR update (applies both to manual and automatic ETR updates).
CalculatedETRSource	String	true	The name of the user or system that initiated the last ETR recalculation (applies to automatic ETR updates only).
AreaNoteETRSource	String	true	The name of the user or system that initiated the last area setting ETR update.
ETACrew	String	false	The list of crews associated with the incident's estimated time of arrival (ETA)
CustomerModifiedTime	String	false	The time when the customer status was modified.
Created	String	false	The time when the incident was created.
Modified	String	false	The time when the incident was last modified.
IsPredictedStable	Boolean	false	Flag for the Predicted Stable status.
IsPredicted	Boolean	false	Flag for the Predicted incident type.
IsTypePredicted	Boolean	false	Flag for the Predicted incident type.
CanConvertIncident	Boolean	false	Flag for the single customer and non-outage incidents that can be converted into a new incident type.
PointOfFailure	String	true	Point of failure.
IsPendingClear	Boolean	false	Flag for the Pending Clear status.
IsTypePredicted	Boolean	false	Flag for the Predicted status.
CurrentCriticalCustomer	Integer	false	Current number of type 1 critical customers.
CurrentCriticalCustomer2	Integer	false	Current number of type 2 critical customers.
CurrentCriticalCustomer3	Integer	false	Current number of type 3 critical customers.
CurrentCriticalCustomer4	Integer	false	Current number of type 4 critical customers.
CurrentCriticalCustomer5	Integer	false	Current number of type 5 critical customers.

Field Name	Type	Nullable	Description
HasHazardousCustomer	Boolean	false	Flag for a hazardous customer associated with the incident.
HasRestorationSteps	Boolean	false	Flag for restoration steps associated with the incident.
SystemContributing	Integer	false	The system that is responsible for the incident.
ContributingCause	Integer	false	The cause of the incident.
FaultLocationSource	Integer	false	The source that identified the incident.
DistributionLocationNumber	Integer	false	Distribution location number.
FaultLatitude	Double	true	Latitude of the fault that is associated with the incident.
FaultLongitude	Double	true	Longitude of the fault that is associated with the incident.
FaultID	String	true	ID of the fault that is associated with the incident.
IsFaultLocationManuallyOverridden	Boolean	false	Specifies whether the fault location was manually set by the user (by dragging a fault symbol from the geographic view to the Update Incident display).
HistoryCorrectionDepartment	String	true	History correction department.
HistoryCorrectionReason	String	true	History correction reason.
HistoryCorrectionApprovedBy	String	true	Name of the person who approved the history correction.
HistoryCorrectionApprovedDate	DateTime	true	Date and time when the history correction was approved.
Remarks	String	true	Operator remarks.
CreatedAfterTheFact	Boolean	false	Flag for historical incidents.
EvaluateCrewName	String	true	Name of the evaluator crew.
IsEvaluatorAssigned	Boolean	false	Flag for the evaluator assigned.
CrewsAssignedCount	String	true	The number of assigned crews.
AcknowledgedTime	DateTime	false	Acknowledged time.
ReceivedCompletionFromWfm	Boolean	false	Received completion from WFM.
IncidentClearAttemptFailed	Boolean	false	Specifies whether incident clearing attempts failed.
AMIOutCount	Integer	false	AMI Out count.
AMIIInCount	Integer	false	AMI In count.
WorkOrderId	String	true	ID of the work order that is associated with the incident.
DurationInStatus	Integer	false	The number of minutes an incident stays in the current working status.
PowerQualityType	Integer	false	The "Power quality type" incident completion code.

Field Name	Type	Nullable	Description
NonOutageType	Integer	false	Type of the non-outage incident (returns a NonOutageType coded translation item). This could be a numeric value of one of the following: None = 0 Isolated = 1 Street Light = 2 Hazard = 3 Momentary = 4 Traffic Light = 5 Pole Maintenance = 6 Wire Down = 7
Weather	Integer	false	The "Weather" incident completion code.
WorkQueues	Integer	false	Work queues group outages according to the type of work needed for power restoration (for example, "tree work"), especially during storms with widespread outages.
SystemAffected	Integer	false	The system type affected by the incident. The value is based on the SystemAffected coded translation item.
FailureMode	Integer	false	The "Failure mode" incident completion code.
Manufacturer	Integer	false	The "Manufacturer" incident completion code.
Animals	Integer	false	The "Animals" incident completion code.
Plants	Integer	false	The "Plants" incident completion code.
Locked	Boolean	false	Specifies whether the incident is locked by a user.
SwitchingSource	Integer	false	The switching source of the first switching step (0 = None, 1 = Commanded). The switching source is Commanded for Confirmed and ManuallyConfirmed incidents, including those created as a result of load shed.
Workstation	String	true	The workstation of the user who locked the incident.
LockingUserName	String	true	Locking user name.
ViewportName	String	true	Locking viewport name.
CallbacksSent	Integer	false	Specifies the number of callbacks sent.
HasRelatedIncidents	Boolean	false	Specifies whether the incident has related incidents (incidents that were merged or rolled up to the incident).
IsPredictedDeviceWrong	Boolean	false	Specifies whether the device is incorrectly predicted out. A crew can report a correct device when updating an incident.
AssociatedAssessmentLocationsCount	Integer	false	The number of the associated damage assessment locations.
HistoryCorrectionApprovedDateOffset	DateTimeOffset	false	History correction approved date and time offset.
OutageIslandIds	String	true	Comma-separated list of incident outage island CRCs.

Field Name	Type	Nullable	Description
TroubleDescription	String	true	Lists aggregated trouble codes for all associated customer calls (for example, trouble code "0.0.0.0.0" is equal to a call with Call Type = None, Condition_1 = None, Condition_2 = None, Description = None, and Priority = None).
TroubleDescriptionsTranslated	String	true	Lists aggregated trouble codes for all associated customer calls (for example, trouble code "0.0.0.0.0" is equal to a call with Call Type = None, Condition_1 = None, Condition_2 = None, Description = None, and Priority = None). For trouble codes that are translated into human-readable names using the TroubleDescription coded translation item, returns short names of the associated coded translation item entries.
DistinctTroubleCodes	Collection	false	The collection of TroubleDescription codes.
HasScheme	Boolean	false	Specifies whether the device of record is part of an automated local peer-to-peer switching scheme.
IsNonOutage	Boolean	false	Specifies whether this is a non-outage incident.
IsNonOutageIncident	Boolean	false	Specifies whether this is a non-outage incident (used in the v2 architecture).
IsOutage	Boolean	false	Specifies whether this is an outage incident.
Locked	Boolean	false	Specifies whether the incident is locked by a user.
LockingUserName	String	true	Locking user name.
MeasName	String	true	Measurement name.
PublicInflictedDamageIndicator	Boolean	false	Specifies whether the incident is inflicted by public.
RelatedIncidentsString	String	true	IDs of related incidents (incidents that were merged or rolled up to the incident).
TotalConnectedKVAInterrupted	Float	false	The total connected power (in KVA) interrupted on the nominal feeder.
TotalConnectedKVAServed	Float	false	The total connected power (in KVA) served on the nominal feeder.
ServiceIndex	Float	false	Service index for the incident. Calculated as a sum of service indexes for all affected customers.
SafetyIndex	Float	false	Safety index for the incident. Calculated as a sum of safety indexes for all affected customers.
BaselineIncidentPriorityIndex	Float	false	Baseline priority index for the incident. For outage incidents, calculated as a sum of safety indexes multiplied by safety index weights and service indexes multiplied by service index weights for all affected customers. For non-outage incidents, the value is fixed and set to the "Baseline Priority Index for Non-Outage Incidents" defined in the OMS Server Configuration Editor.

Field Name	Type	Nullable	Description
AdjustedIncidentPriorityIndex	Float	false	Adjusted priority index for the incident. Calculated as a baseline priority index multiplied by the "restoration urgency", "critical customers", "exact value" and "threshold" multipliers (Area Setting, Device Type, Duration, CMI, etc.) defined in the OMS Server Configuration Editor.
IncidentPriorityIndex	Float	false	Priority index for the incident. If the feeder restoration priority (FRP) value is specified, the priority index is set to Double.MaxValue (1.7976931348623157E+308)/(FRP+1.0); otherwise, the priority index is set to the adjusted priority index.
ViewportName	String	true	Locking viewport name.
WastedTrip	Boolean	false	Specifies whether the incident was not company responsibility (wasted trip).
Workstation	String	true	The workstation of the user who locked the incident.
DailySeqNumber	Integer	false	Daily sequence number.
PingOut	Integer	false	The number of affected customers with the Ping Out status.
PingIn	Integer	false	The number of affected customers with the Ping In status.
PingUndetermined	Integer	false	The number of affected customers with the Ping Undetermined status.
PingPending	Integer	false	The number of affected customers with the Ping Pending status.
PingInProgress	Boolean	false	Specifies whether the affected customer meter pinging is in progress.
PingedCustomers	Integer	false	The number of pinged customers.
PingsSent	Integer	false	The number of ping requests sent from the Customers tab of the Update Incident display.
CallsHaveBeenCleared	Boolean	false	Specifies whether calls have been cleared.
StatusStartTime	DateTime	false	Date and time when the incident was set to the current status.
ManualEtr	DateTime	false	Date and time when the incident ETR was manually set.
JustCrewId	String	true	ID of regular crew (not evaluator crew).
SuppressNotificationReason	Integer	false	Suppress notification type code.
OpenDevicePhaseInd	Boolean	false	Open device has a phase indicator?
DeviceSize	String	true	The open device size.
SwitchOrderNames	String	true	The list of the associated switching order names.
OriginalDeviceNameId	String	true	Original device name and ID.
MergedNonOutageCalls	Collection	false	The list of merged non-outage calls.
CalculateConnectedKVA	Boolean	false	Specifies whether connected load in kVA is calculated.

Field Name	Type	Nullable	Description
FieldConfirmed	Boolean	false	Specifies whether the incident was confirmed from the field.
WfmRestorationTime	DateTime	false	Actual restoration time received from the WFM.
HasSwitchingSteps	Boolean	false	Specifies whether the incident has switching steps.
AOR2	String	true	Incident AOR (used for backward compatibility).
AOR3	String	true	Incident AOR (used for backward compatibility).
AOR4	String	true	Incident AOR (used for backward compatibility).
LinkedOrderIds	String	true	The list of linked order IDs

Navigation properties available for Incidents:

- ChronologyLog
- RestorationSteps
- Customers
- UnassociatedCalls
- SwitchingSteps
- CustomFields

4.2. Crews

<http://localhost:8800/Oms/Crews>

Table 2. Crews

Field Name	Type	Nullable	Description
VehicleCount	Integer	false	Number of vehicles assigned to the crew.
VehicleId	String	true	Vehicle ID of the primary employee of the crew.
Id	String	false	Unique ID of the crew assigned by ADMS.
Name	String	true	Name of crew – free form.
CrewId	String	false	Unique ID of the crew assigned by the customer company.
CrewArea	String	true	The area that the crew operates within.
ParentCrewId	String	true	ID of the parent crew.
CrewTypeName	String	true	Name of the crew type (for example, Local Lineman).
Dept	String	true	Department the crew belongs to.
IsEvaluator	Boolean	false	Indicates if the crew includes an evaluator.
Active	Boolean	false	Indicates if the crew is active.
IsAvailable	Boolean	false	Indicates if the crew is currently available.
DivisionTypeCode	Integer	false	Division type code.

Field Name	Type	Nullable	Description
CrewDivision	String	true	Division the crew belongs to.
Division	String	true	The division that the crew operates within.
OpCenter	String	true	The operating center within a division that the crew reports to.
Headquarters	String	true	Crew headquarters that crew reports to.
DefaultLocationX	Single	false	Default location X.
DefaultLocationY	Single	false	Default location Y.
CurrentLocationX	Single	false	Current location X.
CurrentLocationY	Single	false	Current location Y.
EmployeeCount	Integer	false	Number of employees the crew has.
PrimaryEmployeeName	String	true	Name of the primary employee of the crew.
PhoneNumber	String	true	Phone number of the primary employee.
RadioNumber	String	true	Radio number of the primary employee.
AutomaticDispatchId	String	true	Work force management ID provided to the primary employee of the crew.
CrewStatusCode	Integer	false	Crew status code from the ARMSStatus coded translation item.
CrewWorkStatusCode	String	true	Crew work status code from the CrewWorkStatus coded translation item.
IncidentId	String	true	IDs of all incidents the crew is assigned to.
IncidentCount	Integer	false	The number of incidents assigned to the crew.
CurrentIncident	String	true	Name of the incident the crew is currently working on (optionally can be added from WFM).
ETA	DateTime	false	Estimated time for arrival (ETA).
MDTFlag	Boolean	false	Does crew have a mobile data terminal?
IsTemporary	Boolean	false	Is it a temporary crew?
IsAdhoc	Boolean	false	Indicates if the crew was added locally from the OMS Display plugin.
CreatedBy	String	true	Specifies the name of the user (operator) who created the crew record locally.
IsForeign	Boolean	false	Is this crew a foreign crew?
IsContractor	Boolean	false	Is this crew a contract crew?
AvailableDuration	Integer	false	Time period, in minutes, since the crew was de-assigned from the previous incident (staying idle).
OnJobDuration	Integer	false	Time period, in minutes, since the crew was assigned to the current incident (being work).
CrewAreaId	String	false	Crew area ID.
UserId	String	false	ID of the user who performs a crew modeling action.
AvailableTime	DateTime	false	Time the crew is available.

Field Name	Type	Nullable	Description
Location	String	true	Location.
DefaultCrewAreaID	String	true	ID of the default crew area.
DefaultCrewArea	String	true	Default area the crew is assigned to.
TempArea	String	true	Area the crew is temporarily assigned to.
HomeArea	String	true	Crew home area.
CurrentIncidentName	String	true	Name of the incident that the crew is currently working on. The current incident is determined using the CurrentIncident field; if CurrentIncident is blank, the incident with the highest working status among the incidents assigned to the crew is considered the current incident.
CurrentIncidentDevice	String	true	Device of record for the incident the crew is currently working on.
CurrentIncidentFeeder	String	true	Feeder of the incident that the crew is currently working on.
CurrentIncidentEtr	DateTime	false	ETR of the incident that the crew is currently working on.
CurrentIncidentType	Integer	false	Type of incident that the crew is currently working on.
CurrentIncidentDurationInStatus	Integer	false	Duration of the current status for the incident the crew is currently working on.
CurrentIncidentDuration	Integer	false	Time period, in minutes, since the incident start time for the incident the crew is currently working on.
CurrentIncidentCustomersAffectedCount	Integer	false	Number of affected customers for the incident the crew is currently working on.
CurrentIncidentCallCount	Integer	false	Number of customer calls for the incident the crew is currently working on.
CurrentIncidentCriticalCustomerCount	Integer	false	Number of affected critical customers for the incident the crew is currently working on.
CrewShiftStatusCode	Integer	false	Crew shift status code from the CrewShiftStatus coded translation item.
MobileCommunicationStatusCode	Integer	false	Crew job communications code from the MobileCommunicationStatus coded translation item.
CrewAvailabilityStatusCode	Integer	false	Crew availability status code from the CrewAvailabilityStatus coded translation item.

Navigation property available for Crews:

- Employees
- CustomFields

4.3. Employees

<http://localhost:8800/Oms/Employees>

Table 3. Employees

Field Name	Type	Nullable	Description
Id	String	false	Unique ID of the employee.
EmployeeId	String	true	Company assigned Employee ID.
Name	String	true	Name of the employee.
NameId	String	true	Name and ID of the employee in the format <Name> (<ID>).
FirstName	String	true	First name of the employee.
MiddleName	String	true	Middle name of the employee.
LastName	String	true	Last name of the employee.
Prefix	String	true	The employee's phone prefix.
Suffix	String	true	The employee's phone suffix.
PhoneNumber	String	true	Phone number of the employee.
SecondaryPhoneNumber	String	true	Secondary phone number of the employee.
RadioNumber	String	true	Radio number of the employee.
DefaultRadioNumber	String	true	Default radio number of the employee.
AutomaticDispatchId	String	true	Automatic dispatch ID provided by the crew management system.
DefaultAutomaticDispatchId	String	true	Default automatic dispatch ID provided by the crew management system.
DefaultCrewArea	String	true	Default area the crew assigned to.
TempArea	String	true	Area the crew is temporarily assigned to.
CrewArea	String	true	Crew area.
CrewAreaID	String	true	ID of the crew area.
DefaultCrewAreaID	String	true	ID of the default crew area.
SupervisorID	String	true	ID of the employee's supervisor.
SupervisorName	String	true	Name of the employee's supervisor.
DefaultLeadID	String	true	Name of the employee's default lead.
CurrentLeadID	String	true	Name of the employee's current lead.
EmployeeTypeID	String	true	ID of the employee type.
EmployeeTypeName	String	true	Name of the employee type.
Dept	String	true	Department the employee belongs to.
DeptID	String	true	ID of the department.
VehicleTypeID	String	true	ID of the vehicle type.
CrewDivision	String	true	Division the employee crew belongs to.

Field Name	Type	Nullable	Description
IsEvaluator	Boolean	false	Is the employee an evaluator?
IsExcluded	Boolean	false	Is employee excluded?
MembersCount	Integer	false	Number of members that are associated with this employee.
Division	String	true	Division the employee belongs to.
OpCenter	String	true	Operating center the employee belongs to.
ShoeSize	String	true	Shoe size of the employee.
IsContractor	Boolean	false	Is the employee a contractor?
CapabilityCode	Integer	false	Numerical value of the capability code as configured by the analyst.
CanSwitch	Boolean	false	Can this employee do the switching operation?
IsAvailable	Boolean	false	Is the employee available based on the shift rules?
EmployeeStatusCode	Integer	false	Numerical value of the employee status code as configured by the analyst.
EMPHQ	String	true	Headquarter of the employee.
CrewName	String	true	Name of the crew this employee is associated with.
CrewId	String	true	ID of the crew this employee is associated with.
TempCrewName	String	true	Name of the temporary crew this employee is associated with.
IsPrimary	Boolean	false	Is this employee a primary employee?
IsAdhoc	Boolean	false	Indicates if the employee was added locally from the OMS Display plugin.
CreatedBy	String	true	Specifies the name of the user (operator) who created the employee record locally.
VehicleTypeText	String	true	The vehicle type that this employee has.
VehicleTypeCode	Integer	false	Numerical value of the vehicle type code as configured by the analyst
VehicleIDs	String	true	IDs of the vehicles assigned to this employee.
MDTFlag	Boolean	false	Does this employee have a mobile data terminal?
IsForeign	Boolean	false	Is the employee a foreign employee?
DefaultLeadName	String	true	The name of the default lead of this employee.
CurrentLeadName	String	true	The name of the current lead of this employee.
UserId	String	false	ID of the user who performs an employee modeling action.

4.4. PlannedOutages

<http://localhost:8800/Oms/PlannedOutages>

Table 4. PlannedOutages

Field Name	Type	Nullable	Description
Id	String	false	Unique ID of the planned outage.
Name	String	true	Human readable name.
CreatedTime	DateTime	false	Creation time of the planned outage.
PrimaryStartTime	DateTime	false	Primary start time of the planned outage.
PrimaryEndTime	DateTime	false	Primary end time of the planned outage.
AlternateStartTime	DateTime	false	Alternate start time of the planned outage.
AlternateEndTime	DateTime	false	Alternate end time of the planned outage.
UseAlternateDate	Boolean	false	Indicates whether alternate date is used for the planned outage.
IsStartTimeExpired	Boolean	false	Indicates whether the planned outage start time expired.
IsStartTimeExpiring	Boolean	false	Indicates whether the planned outage start time is expiring.
EventType	String	true	Type of the event for which the planned outage is scheduled.
CausesIncident	Boolean	false	Event will create an outage so an incident will be created when the start time is due.
DeviceId	String	true	ID of the device that will be operated to create the planned outage.
DeviceName	String	true	Name of the device that is operated to create the planned outage.
DevicePtX	Single	false	X-coordinate of the planned outage device.
DevicePtY	Single	false	Y-coordinate of the planned outage device.
DeviceType	Integer	false	Type of the planned outage device.
DeviceTypeString	String	true	DNOM type of the planned outage device (for example, dmsSwitch).
SwitchType	Integer	false	Switch type of the planned outage device. Switch types can be the following: 0: None 1: Switch 2: Fuse 3: Recloser 4: Breaker 5: Sectionalizer 6: Jumper 7: Network Protector
HasHazardousCustomer	Boolean	false	Flag for a hazardous customer associated with the planned outage.
CustomerAccountsString	String	true	List of customer account numbers affected by the planned outage.

Field Name	Type	Nullable	Description
CustomerAccountsNotifyString	String	true	List of pairs of affected customer account numbers and notification statuses. For example, 555110155555110155555110155= True,55537452055374520555374520= False.
AddedCustomersString	String	true	List of customer account numbers added to the planned outage affected customers.
AddedCustomersNotifyString	String	true	List of pairs of added customer account numbers and notification statuses.
RemovedCustomersString	String	true	List of customer account numbers removed from the planned outage affected customers.
RemovedCustomersNotifyString	String	true	List of pairs of removed customer account numbers and notification statuses.
Station	String	true	Name of the station planned outage is part of.
OpCenter	String	true	Service center that has the planned outage.
ServiceCenter	String	true	ID of the service center that has the planned outage.
ServiceCenterName	String	true	Name of the service center that has the planned outage.
Division	String	true	Division that has the planned outage.
RepresentativeName	String	true	Name of the representative to talk to about the planned outage.
RepresentativeNumber	String	true	Telephone number of the representative to talk to about the planned outage.
RepresentativeEmail	String	true	Email address of the representative to talk to about the planned outage.
RepresentativeAddress	String	true	Address of the representative to talk to about the planned outage.
OutageMessage	String	true	Outage message that was sent as part of the planned outage.
CreatorComments	String	true	Comments from the planned outage creator.
AOR	String	true	Name of the Area of Responsibility (AOR) the planned outage belongs to.
SwitchOrderId	Collection	true	A collection of switching order IDs that are associated with the planned outage.
SwitchOrderName	Collection	true	A collection of switching order names that are associated with the planned outage.
SwitchOrderNames	String	true	A list of switching order names that are associated with the planned outage.
Created	DateTime	false	The time when the incident was created.
Modified	DateTime	false	The time when the incident was last modified.
FeederName	String	true	Feeder name.
WorkOrderId	String	true	Work Order ID.

Field Name	Type	Nullable	Description
PlannedOutageType	Integer	false	Planned outage type from the PlannedOutageType coded translation item.
PlannedIncidentId	String	true	ID of the incident associated with the planned outage.
AlarmSent	Boolean	false	Indicates whether an alarm about the start time expiration was already issued (the alarm can be issued only once).
IsEmergent	Boolean	false	Indicates whether this is an emergent planned outage. This flag is based on the number of hours from the planned outage creation time until the planned outage start time, as configured in the OMS Server Configuration Editor.
Status	Integer	false	Planned outage status (for example, 0 = Active or 1 = Cancelled). The statuses are defined by the PlannedOutageStatus coded translation item.
LastStatusChangeDateTimeUtc	DateTime	false	The date and time of the last status update.
FieldName	String	true	Custom field name.

Navigation property available for PlannedOutages:

- CustomFields

4.5. Customers

<http://localhost:8800/Oms/Customers>

Table 5. Customers

Field Name	Type	Nullable	Description
CustIdx	Integer	false	Customer index for faster access.
CustomerKey	String	true	The unique key of the customer.
ServicePoint	String	true	The transformer ID the customer belongs to. The field format is SP_<Transformer ID>.
MeterNumber	String	true	The customer's meter number.
AMROutageReport	Boolean	false	Does this customer have AMR outage message?
AMRPowerOnReport	Boolean	false	Does this customer have AMR power on message?
PhoneCall	Boolean	false	Has this customer called?
PriorityCall	Boolean	false	Indicates whether this customer has a priority call, as configured on the "Is a Priority Call" tab in the OMS Server Configuration Editor.
X	Single	false	X-coordinate of customer location.
Y	Single	false	Y-coordinate of customer location.
CallType	Integer	false	Numerical value of the customer call type.

Field Name	Type	Nullable	Description
Condition_1	Integer	false	Numerical value of the first condition specified in the customer call.
Condition_2	Integer	false	Numerical value of the second condition specified in the customer call.
Description	Integer	false	Numerical value of the description specified in the customer call, as configured by the analyst.
PriorityCode	Integer	false	Numerical value of the call priority type.
AccountNumber	String	true	Account number of the customer.
Address	String	true	Address of the customer.
AreaCode	String	true	Area code of customer's phone number.
Comment	String	true	Any additional comment provided by the customer.
CriticalAccount	String	true	The key of the customer's critical type 1.
CriticalAccount2	String	true	The key of the customer's critical type 2.
CriticalAccount3	String	true	The key of the customer's critical type 3.
CriticalAccount4	String	true	The key of the customer's critical type 4.
CriticalAccount5	String	true	The key of the customer's critical type 5.
CustomerName	String	true	The customer's full name.
IncidentID	String	true	Incident ID this customer is associated with.
PhonePrefix	String	true	The customer's phone prefix.
PhoneSuffix	String	true	The customer's phone suffix.
LastAMRReportTime	DateTime	false	The time of the last AMR report for this customer.
LastCallTime	DateTime	false	The time of the last call from this customer.
EnergizedTime	DateTime	false	The time when the customer was restored.
Modified	DateTime	false	The time when the customer status was modified.
OperatingCenter	String	true	Operating center the customer belongs to.
Division	String	true	Division the customer belongs to.
Column1-12(*)	String	true	These 12 fields may contain additional customer data.
VoltA	String	true	Phase A voltage on a customer meter in Volts.
VoltB	String	true	Phase B voltage on a customer meter in Volts.
VoltC	String	true	Phase C voltage on a customer meter in Volts.

Note

To show additional customer data (for example, load ID and feeder ID), add the field names as they appear in the customer schema file (LoadId and Feeder) to the CUSTOMERFILE_CUSTOM_COLUMNS parameter in the Net Data Server configuration file. For example, add "CUSTOMERFILE_CUSTOM_COLUMNS = LoadId,Feeder".

Navigation property available for customers:

- CustomerHistory

4.6. HierarchyGroups

<http://localhost:8800/Oms/HierarchyGroups>

Table 5. HierarchyGroups

Field Name	Type	Nullable	Description
GroupType	String	true	Type of the hierarchy group.
Name	String	true	Name of the hierarchy group.
ID	String	true	ID of the hierarchy group.
GroupId	String	true	ID and type of the hierarchy group.
AOR	String	true	Area of responsibility for the hierarchy group.
StationId	String	true	Station that the hierarchy group is associated with.
FeederId	String	true	Feeder that the hierarchy group is associated with.
ParentName	String	true	Parent name of the hierarchy group.
ParentId	String	true	Parent ID of the hierarchy group.
GrandParentId	String	true	Grandparent ID of the hierarchy group.
GreatGrandParentId	String	true	Great grandparent ID of the hierarchy group.
up1GroupName	String	true	Level 1 parent name of the hierarchy group.
up2GroupName	String	true	Level 2 parent name of the hierarchy group.
up3GroupName	String	true	Level 3 parent name of the hierarchy group.
up4GroupName	String	true	Level 4 parent name of the hierarchy group.
up5GroupName	String	true	Level 5 parent name of the hierarchy group.
up6GroupName	String	true	Level 6 parent name of the hierarchy group.
up7GroupName	String	true	Level 7 parent name of the hierarchy group.
up8GroupName	String	true	Level 8 parent name of the hierarchy group.
up9GroupName	String	true	Level 9 parent name of the hierarchy group.
up10GroupName	String	true	Level 10 parent name of the hierarchy group.
up1GroupId	String	true	Level 1 parent ID of the hierarchy group.
up2GroupId	String	true	Level 2 parent ID of the hierarchy group.
up3GroupId	String	true	Level 3 parent ID of the hierarchy group.
up4GroupId	String	true	Level 4 parent ID of the hierarchy group.
up5GroupId	String	true	Level 5 parent ID of the hierarchy group.
up6GroupId	String	true	Level 6 parent ID of the hierarchy group.
up7GroupId	String	true	Level 7 parent ID of the hierarchy group.
up8GroupId	String	true	Level 8 parent ID of the hierarchy group.
up9GroupId	String	true	Level 9 parent ID of the hierarchy group.

Field Name	Type	Nullable	Description
up10GroupId	String	true	Level 10 parent ID of the hierarchy group.
CustomerCount	Integer	false	The number of customers.
ServicePointCount	Integer	false	The number of service points.
X	Single	false	X-coordinate of the incident.
Y	Single	false	Y-coordinate of the incident.
IncidentsUnassigned	Integer	false	Number of unassigned incidents for this hour.
IncidentsAssigned	Integer	false	Number of assigned incidents for this hour.
IncidentsDispatched	Integer	false	Number of incidents for which crews have been dispatched.
IncidentsEnroute	Integer	false	Number of incidents for which crews are en route.
IncidentsOnSite	Integer	false	Number of incidents for which crews are onsite.
IncidentsComplete	Integer	false	Number of incidents that are in complete stage.
IncidentCount	Integer	false	The number of incidents.
IncidentCurrentCustomersOut	Integer	false	The number of customers that are currently de-energized due to incidents.
IncCountTypeNone	Integer	false	Number of incidents where the device type is not specified.
IncCountTypeCut	Integer	false	Number of incidents where the device type is a cut.
IncCountTypeFuse	Integer	false	Number of incidents where the device type is a fuse.
IncCountTypeJumper	Integer	false	Number of incidents where the device type is a jumper.
IncCountTypeLineSwitch	Integer	false	Number of incidents where the device type is a switch.
IncCountTypePoleRecloser	Integer	false	Number of incidents where the device type is a recloser.
IncCountTypeSectionalizer	Integer	false	Number of incidents where the device type is a sectionalizer.
IncCountTypeStationBreaker	Integer	false	Number of incidents where the device type is a breaker.
IncCountTypeTransformer	Integer	false	Number of incidents where the device type is a transformer.
IncCountTypeNetworkProtector	Integer	false	Number of incidents where the device type is a network protector.
IncCountTypeServicePoint	Integer	false	Number of incidents where the device type is a service point.
IncCountTypeTransformerBank	Integer	false	Number of incidents where the device type is a transformer bank.
MomentaryOutageCount	Integer	false	The number of momentary incidents.
ConfirmedOutageCount	Integer	false	The number of confirmed incidents.
PredictedOutageCount	Integer	false	The number of predicted incidents.

Field Name	Type	Nullable	Description
SingleCustomerOutageCount	Integer	false	The number of single customer incidents.
NonOutageCount	Integer	false	The number of non-outage incidents.
ActiveIncidents	Integer	false	The number of active incidents.
NewIncidents	Integer	false	Number of new incidents in this hour.
CompletedIncidents	Integer	false	The number of completed incidents.
DamageCount1	Integer	false	Number of poles down.
DamageCount2	Integer	false	Number of spans of wire down.
DamageCount3	Integer	false	Number of transformers affected.
DamageCount4	Integer	false	Number of services down.
DamageCount5	Integer	false	Number of cross arms down.
DamageCount6	Integer	false	Number of tree trim spans.
LatestETR	DateTime	false	Latest ETR of the incidents.
AssignedCrewCount	Integer	false	The number of assigned crews.
CrewNeededCount	Integer	false	The number of Line Crew Needed flags.
TreeCrewNeededCount	Integer	false	The number of Tree Crew Needed flags.
FirstResponderNeededCount	Integer	false	The number of First Responder Needed flags.
EvaluatorNeededCount	Integer	false	The number of Evaluator Needed flags.
CircuitsOut	Integer	false	The number of circuits out in the current hour.
CrewsAssigned	Integer	false	The number of crews assigned.
CrewsDispatched	Integer	false	The number of crews dispatched.
CrewsEnRoute	Integer	false	The number of crews en route.
CrewsOnBreak	Integer	false	The number of crews on break.
CrewsOnSite	Integer	false	The number of crews on-site.
CrewsOutOfCoverage	Integer	false	The number of crews out of coverage.
CrewsReleased	Integer	false	The number of crews released.
CrewsSignOff	Integer	false	The number of crews with Sign Off status.
CrewsSignOn	Integer	false	The number of crews with Sign On status.
CrewsStandBy	Integer	false	The number of crews standby.
PredictedOutServicePointCount	Integer	false	The number of service points predicted out.
ConfirmedOutServicePointCount	Integer	false	The number of service points confirmed out.
TotalCustomerCount	Integer	false	The total number of customers.
PeakCustomersOut	Integer	false	Peak customers out during this hour.
NewCustomersOut	Integer	false	Number of new customers out this hour.
CustomersRestored	Integer	false	The number of customers restored.
PriorityCustomersOut	Integer	false	The number of de-energized customers that sent high or extreme call priority type calls.

Field Name	Type	Nullable	Description
CriticalCustomersOut	Integer	false	The number of critical customers out.
CallAggregation1-10	Integer	false	The number of calls satisfying each of the call aggregation rules.
ActiveCalls	Integer	false	The number of active calls.
ActiveAmis	Integer	false	Number of active AMI In/Out messages.
ActiveEvents	Integer	false	The number of active events.
CriticalCustomerEvents	Integer	false	The number of critical customer events.
PriorityCustomerEvents	Integer	false	The number of priority calls, as configured on the "Is a Priority Call" tab in the OMS Server Configuration Editor.
MomentaryEventCustomers	Integer	false	The number of customers in momentary events.
MomentaryIncidentCustomers	Integer	false	The number of customers in momentary incidents.
MomentaryOutCustomerCount	Integer	false	The number of customers de-energized due to momentary incidents.
ConfirmedOutCustomerCount	Integer	false	The number of customers that are confirmed to be de-energized.
ConfirmedInCustomerCount	Integer	false	The number of customers that are confirmed to be energized.
PredictedOutCustomerCount	Integer	false	The number of customers that are predicted to be de-energized.
TotalAffectedCustomerCount	Integer	false	The total number of affected customers.
TotalOutCustomerCount	Integer	false	The total number of de-energized customers.
CMI	Double	false	CMI of the incidents.
SAIDI	Double	false	SAIDI value for this hour.
SAIFI	Double	false	SAIFI value of this hour.
MAIFI	Double	false	MAIFI value of this hour.
MAIFIE	Double	false	MAIFIE value of this hour.
AssignedEvaluateCrewCount	Integer	false	The number of assigned evaluators.

Navigation properties available for HierarchyGroups:

- Parent
- ChildHierarchyGroups
- RecentHistoryEntries
- CurrentHistory

Available service operations:

- GetOpCenters() returns a comma separated string of OpCenter names associated with the incidents in the system.
- GetSubstations() returns a comma separated string of Substation names associated with the

incidents in the system.

4.7. RestorationSteps

<http://localhost:8800/Oms/RestorationSteps>

Table 6. RestorationSteps

Field Name	Type	Nullable	Description
Id	String	false	Unique ID of the restoration step.
IncidentId	String	true	Incident ID this restoration step is associated with.
StepNum	Integer	false	Step number of the restoration.
Station	String	true	Name of the station the device belongs to.
DeviceId	string	true	ID of the device the restoration step is associated with.
DeviceName	String	true	Name of the device the restoration step is associated with.
CloseDeviceId	String	true	ID of the device that closed this restoration step.
CloseDeviceName	String	true	Name of the device that closed this restoration step.
CustomerAccountNumbersString	String	true	A list of the associated customer account numbers.
CustomerKeysString	String	true	A list of the associated customer key numbers.
CustomerCount	Integer	false	Number of customers affected/restored by the restoration step.
StepCmi	Integer	false	CMI of the restoration step.
MomentaryCount	Integer	false	Momentary count.
DeenergizingDeviceFeederName	String	true	Name of the feeder the de-energizing device belongs to.
DeenergizingDeviceServiceCenterId	String	true	ID of the service center the de-energizing device belongs to.
DeenergizingDeviceServiceCenterName	String	true	Name of the service center the de-energizing device belongs to.
EnergizingDeviceType	String	true	Name of the energizing device type.
EnergizingSwitchTypeCode	Integer	false	Type code of the energizing switch.
EnergizingSwitchType	String	true	Type of the energizing switch.
EnergizingDeviceFeederName	String	true	Name of the feeder the energizing device belongs to.
EnergizingDeviceServiceCenterName	String	true	Name of the service center the energizing device belongs to.
EnergizingDeviceServiceCenterId	String	true	ID of the service center the energizing device belongs to.

Field Name	Type	Nullable	Description
EnergizingDeviceTypeCode	Integer	true	Type code of the energizing device.
Phase	String	false	Phases enumeration of the energizing device.
EnergizedTime	DateTime	false	Time when the switch was energized to close the step.
DeenergizedTime	DateTime	false	Time when the restoration step was created.
CancelRestorationStep	Boolean	false	Do not include in the statistics calculation.
ExcludeFromCustomerCount	Boolean	false	Do not include in the customer count calculation.
DeenergizingDeviceType	String	true	Type of the de-energizing device.
DeenergizingDeviceTypeCode	Integer	false	Type code of the de-energizing device.
DeenergizingSwitchType	String	true	Type of the de-energizing switch
DeenergizingSwitchTypeCode	Integer	false	Type code of the de-energizing switch.
DeenergizingDeviceIsScada	Boolean	false	De-energizing device is SCADA.
EnergizingDeviceIsScada	Boolean	false	Energizing device is SCADA.
EnergizingDeviceStationName	String	true	Station name of the energizing device.

4.8. RestorationStepSummaries

<http://localhost:8800/Oms/RestorationStepSummaries>

Table 7. RestorationStepSummaries

Field Name	Type	Nullable	Description
Id	String	false	Unique ID of the restoration step summary.
IncidentId	String	true	Incident ID this restoration step is associated with.
StepNum	Integer	false	Step number of the restoration.
RestorationStepId	String	true	Unique ID of the restoration step.

4.9. SwitchingSteps

<http://localhost:8800/Oms/SwitchingSteps>

Table 8. SwitchingSteps

Field Name	Type	Nullable	Description
StepNum	Integer	false	Step number of the restoration.
Station	String	true	Name of the station the device belongs to.
DeviceId	string	true	ID of the device the restoration step is associated with.
DeviceName	String	true	Name of the device the restoration step is associated with.

Field Name	Type	Nullable	Description
AffectedCustomerCount	Integer	false	Number of customers affected by the switching step.
RestoredCustomerCount	Integer	false	Number of customers restored by the switching step.
Phase	String	true	Phases enumeration of the energizing device.
IncidentGuid	Guid	false	Guid of the incident.
SwitchOrderId	string	true	ID of the SwitchOrder.
SwitchOrderStepGuid	Guid	false	Guid of the SwitchOrder step.
ServiceCenter	String	true	ID of the service center the incident belongs to.
ServiceCenterName	String	true	Name of the service center the incident belongs to.
x	Float	false	X-coordinate.
y	Float	false	Y-coordinate.
Feeder	String	true	Feeder.
DeviceType	String	true	Device type.
DeviceTypeCode	Integer	false	Device type code.
SwitchType	String	true	Switch type.
SwitchTypeCode	Integer	false	Switch type code.
Action	String	true	Action.
Operator	String	true	Operator.
CrewName	String	true	Crew name.
OperationTime	DateTime	false	Operation time.
DeviceIsScada	Boolean	false	Device is SCADA.

4.10. ChronologyEntries

<http://localhost:8800/Oms/ChronologyEntries>

Table 9. ChronologyEntries

Field Name	Type	Nullable	Description
Id	String	false	The date and time when the OMS data was last modified.
Time	DateTime	false	Timestamp of the event.
OpCenter	String	true	Name of the service center where the event took place.
ServiceCenterName	String	true	Name of the service center where the event took place.
ServiceCenter	String	true	ID of the service center where the event took place.
Station	String	true	Station name the event belongs to.
Feeder	String	true	Name of the associated feeder.
IncidentName	String	true	Incident number.
UserName	String	true	Name of the user who performed the action.

Field Name	Type	Nullable	Description
CrewName	String	true	Name of the assigned crew.
Action	String	true	Action details.
MsgType	String	true	Type of the chronology log message.
IncidentGuid	String	true	Incident Guid.

4.11. Polygons

<http://localhost:8800/Oms/Polygons>

Table 10. Polygons

Field Name	Type	Nullable	Description
Name	String	false	Name of the polygon object.
Object	String	true	Object associated with the polygon.
ObjectId	String	true	Object ID.
Points	String	true	X and Y coordinates of polygon points.
CenterX	Float	false	X-coordinate of the polygon's center point.
CenterY	Float	false	Y-coordinate of the polygon's center point.
GroupName	String	true	Polygon group name, such as Roads or Zip code.

4.12. AreaNotes

<http://localhost:8800/Oms/AreaNotes>

Table 11. AreaNotes

Field Name	Type	Nullable	Description
Id	String	false	Service center ID.
Name	String	false	Service center name.
SystemMode	String	true	System mode (for example, storm).
OrgUnitId	String	true	Organization hierarchy unit ID from the station or customer model.
IsDeleted	Boolean	false	Indicates if the area setting is deleted.
IsUpdated	Boolean	false	Indicates if the area setting is updated.
ETR	DateTime	false	Estimated time of restoration.
CaseNoteCode	Integer	false	Case note code, according to the CaseNote coded translation item.
AreaNoteText	String	true	Area note text value, according to the AreaNote coded translation item.
LastUpdateTime	DateTime	false	Time when the case note was last modified.

Field Name	Type	Nullable	Description
LastUpdateByUserId	String	true	ID of the user who last updated the case note.
CreationTime	DateTime	false	Time when the case note was created.
FreeFormRemarks	String	true	Area note custom text value specified in the Use Message field on the Update Area Settings display.
OverrideCaseNote	Boolean	false	Indicates if the area setting has a specific area note defined by the Area Note field on the Update Area Settings display.
ExcludeCMI	Boolean	false	Specifies whether the CMI calculation is paused for incidents in the area until the area setting is expired or removed.
PSTDelta	Integer	false	Specifies the percentage value (increment or decrement) to adjust the predicted stable timeout.
PSTDeltaKey	Integer	false	The PSTDelta coded translation item key for the incident's predicted stable timer adjustment.
ExpirationTime	DateTime	false	Area setting expiration time.
DisableAutoRestorationPings	Boolean	false	Specifies whether automatic restoration pings are disabled for customers that are affected by incidents in the area.
ReferenceId	String	true	The work order number that is associated with the storm. (Does not relate to the incident work order number and does not apply to incidents.)
UseFRP	Boolean	false	Specifies whether the Feeder Restoration Priority (FRP) is used as the order priority for breaker incidents in the area.
RestorationUrgency	Boolean	false	Specifies whether there is restoration urgency and the area setting applies the overall area multiplier to change incident rankings based on the operating mode.
HasStarted	Boolean	false	Specifies whether the area setting has started.
HasExpired	Boolean	false	Specifies whether the area setting has expired.
AutoClearMomentaries	Boolean	false	Specifies whether automatic momentary incident clearing is enabled in the area affected by the area setting.
DisableAutoCrewAssignment	Boolean	false	Specifies whether automatic crew assignment by the Process Automation application is disabled for orders in the area affected by the area setting.
DisableAutoCrewTransfer	Boolean	false	Specifies whether automatic organization transferring by the Process Automation application is disabled for orders in the area affected by the area setting.

Navigation property available for AreaNotes:

- CustomFields

4.13. UnassociatedCalls

<http://localhost:8800/Oms/UnassociatedCalls>

Table 12. Calls

Field Name	Type	Nullable	Description
Id	String	false	ID of the customer call.
CallTime	DateTime	false	Time of the customer call.
Region1	String	true	Deprecated. Returns null.
Region2	String	true	Service center that is associated with the customer call. Used for compatibility with version 3.7 and earlier.
ServiceCenter	String	true	ID of the service center the customer belongs to.
ServiceCenterName	String	true	Name of the service center the customer belongs to.
ServicePoint	String	true	Service point the customer belongs to.
MeterNumber	String	true	Number of the customer's meter.
AccountNumber	String	true	Customer account number.
Phase	String	true	Customer phasing.
CustomerKey	String	true	Customer key.
CustomerName	String	true	Name of the customer.
Address	String	true	Address of the customer.
PhoneNumber	String	true	Phone number of the customer.
BusinessName	String	true	Business name of the customer.
AlternateName	String	true	Alternate contact name for callback.
AlternateContact	String	true	The alternate customer's contact.
Longitude	Single	false	Longitude of the customer's location.
Latitude	Single	false	Latitude of the customer's location.
X	Single	false	X-coordinate of the customer's location.
Y	Single	false	Y-coordinate of the customer's location.
CriticalAccount	String	true	The key of the customer's critical type 1.
CriticalAccount2	String	true	The key of the customer's critical type 2.
CriticalAccount3	String	true	The key of the customer's critical type 3.
CriticalAccount4	String	true	The key of the customer's critical type 4.
CriticalAccount5	String	true	The key of the customer's critical type 5.
SpecialAccount	Integer	false	Special account number of the customer.
CustomerTypeId	String	true	Customer type (returns a CustomerType coded translation item).
CallType	Integer	false	Type of the customer call.
Condition1	Integer	false	Value of the condition 1.
Condition2	Integer	false	Value of the condition 2.
Description	Integer	false	Description of the customer call.
Priority	Integer	false	Call priority type assigned to the customer call.

Field Name	Type	Nullable	Description
CustomTroubleCode	Integer	false	The aggregated trouble codes for the associated customer call (for example, trouble code "3.0.0.0.0" is equal to a call with Call Type = Area Out, Condition_1 = None, Condition_2 = None, Description = None, and Priority = None).
DrivingDirections	String	true	Driving directions.
Comment	String	true	Comment to the customer call.
Informative	Boolean	false	Indicates if the call contains an informative comment from the customer.
PowerOut	Boolean	false	Indicates if the customer is de-energized based on AMI measurements.
EnrgzState	Integer	false	Energizing state.
IsCancelledCall	Boolean	false	Indicates if the customer call is cancelled.
IsCallback	Boolean	false	Indicates if this is a callback call.
WantsCallback	Boolean	false	Indicates if the customer wants callback.
CallbackNumber	String	true	Customer callback number.
CallbackName	String	true	Contact name for callback.
CallDuration	Integer	false	Duration of the call.
LatestCallbackTime	String	false	The time after which the customer does not wish to receive a callback. By default, 12:00 AM.
CancelCallbackRequest	Boolean	false	Specifies whether a callback request has been cancelled.
UserName	String	true	Agent who received the call (for example, Meter Event).
CallSource	String	true	Name of the system that received the trouble call (for example, ADMS or Call Entry).
CallSourceType	String	true	The type of call source (for example, call, text, or meter).
SourceName	String	true	Source name.
IsConvertCall	Boolean	false	Indicates if this is a converted call. Applies to non-outage calls only.
IsNonOutageCall	String	false	Indicates if this is a non-outage call. Applies to non-outage calls only.
IsForwardedCall	String	false	This flag is set in the Call server to indicate whether this call was forwarded from the Net server.
SourceTime	String	false	Source time.
CustomerCallsCount	Integer	false	Number of calls received from the customer in an incident.
IncidentID	String	true	The incident ID this call is associated with.
Street	String	true	The street this call is associated with.
City	String	true	The city this call is associated with.
State	String	true	The state this call is associated with.
ZipCode	String	true	The zip code this call is associated with.

Field Name	Type	Nullable	Description
TroubleDescription	String	true	The aggregated trouble code for the associated customer call (for example, trouble code "3.0.0.0.0" is equal to a call with Call Type = Area Out, Condition_1 = None, Condition_2 = None, Description = None, and Priority = None). For trouble codes that are translated into human-readable names using the TroubleDescription coded translation item, returns short names of the associated coded translation item entries.
RelatedIncidentId	String	true	ID of the associated incident
PhoneCall	Boolean	false	Indicates if this is a phone call.
AMROutageReport	Boolean	false	Indicates if this customer has AMR outage message.
AMRPowerOnReport	Boolean	false	Indicates if this customer has AMR power on message.
AMIPingPending	Boolean	false	Indicates if the AMI ping is pending.
AMIPingOut	Boolean	false	Indicates if the AMI ping is out.
AMIPingIn	Boolean	false	Indicates if the AMI ping is in.
ConfirmedOut	Boolean	false	Indicates if the customer is confirmed out.
IsSpecialAccount	Boolean	false	Indicates if the customer has special account.
IsSpecialAccountSpecified	Boolean	false	Indicates if the customer has certified special account.
CallsCount	Integer	false	The count of customer calls
AreaCode	String	false	Area code.
PhonePrefix	String	false	Phone prefix.
PhoneSuffix	String	false	Phone suffix.
PriorityCall	Boolean	false	Indicates if this is a priority call, as configured on the "Is a Priority Call" tab in the OMS Server Configuration Editor.
CallId	String	false	Call ID.

4.14. LastUpdate

<http://localhost:8800/Oms/LastUpdate>

A timestamp that can be used to determine if updating client data is necessary

Table 13. LastUpdate

Field Name	Type	Nullable	Description
LastUpdate	DateTime	false	The date and time when the OMS data was last modified.

4.15. Orders

<http://localhost:8800/Oms/Orders>

Table 14. Orders

Field Name	Type	Nullable	Description
Id	String	false	ID of the follow-up work order.
Name	String	true	Name of the follow-up work order. The name format is <AssociatedIncidentName>-<FollowUpWorkOrderNumber>.
Comment	String	true	Comment for the follow-up work order.
DeviceName	String	true	Name of the device of record.
DeviceId	String	true	ID of the device of record.
Aor	String	true	Name of the Area of Responsibility (AOR).
WorkOrg	String	true	Name of the assigned work organization, if assigned.
WorkOrg2	String	true	Name of the second assigned work organization, if assigned.
WorkOrg3	String	true	Name of the third assigned work organization, if assigned.
Organization	Integer	false	Name of the organization that the follow-up work order is assigned to.
OrderType	Integer	false	The follow-up work order type.
Priority	Integer	false	Priority of the follow-up work order.
OrderStatus	Integer	false	Status of the follow-up work order.
DeviceStation	String	true	Station of the device of record.
DeviceType	String	true	Type of the device of record.
DeviceFeederName	String	true	Name of the feeder that the device of record is associated with.
DevicePhases	String	true	Phases of the device of record.
CreatedBy	String	true	Name of the user who created the follow-up work order.
ModifiedBy	String	true	Name of the user who modified the follow-up work order
IncidentId	String	true	ID of the incident that the follow-up work order is associated with.
IncidentName	String	true	Name of the incident that the follow-up work order is associated with.
Created	DateTime	false	Date and time when the follow-up work order was created.
Modified	DateTime	false	Date and time when the follow-up work order was last modified.

4.16. CustomFields

URI format:

```
http://localhost:8800/Oms/<ServiceType>('ObjectID')/CustomFields
```

Where:

- **Service Type:** Service type that supports the CustomFields navigation property (Incidents, Crews, PlannedOutages, or AreaNotes).
- **ObjectID:** The ID of the object to provide the list of custom fields for. For example, use the crew ID when querying the Crews service type.

URI example:

```
http://localhost:8800/Oms/Crews('FI1564')/CustomFields?$format=json
```

Table 15. CustomFields

Field Name	Type	Nullable	Description
Key	String	true	Key of the custom field.
Value	String	false	Value of the custom field.

5. Supported GET Functions

Net Data Server (OMS Data Service) supports the following GET functions:

- ActiveIncidents
- CompletedIncidents
- GetCount

Parameter = "entity"

- Statistics
- GetDivisions
- GetOpCenters
- GetSubstations
- GetCustomers

Parameter = "where"

- GetUpdatedCustomers

Parameter = "when"

- GetDeenergizedCustomers

Parameter = "when"

- GetCustomerCalls

Note

The KEEP_CUSTOMER_CALL_HISTORY parameter must be set to true in NetDataServerConfig.txt for the GetCustomerCalls query to show data.

5.1. ActiveIncidents

The "ActiveIncidents" operator can be used to query data for currently active incidents. For example, the following URI returns the data for canceled or not cleared incidents:

```
http://localhost:8800/OMS/ActiveIncidents
```

5.2. Getting the Historical Incident Count

The "GetCount" operator can be used to get the number of cleared and canceled incidents. Currently, the operator supports the HISTORYCORRECTION and FOLLOWUPREQUIRED entities. For example, the URI:

```
http://localhost:8800/OMS/GetCount?Entity='FOLLOWUPREQUIRED'
```

returns the number of cleared and canceled incidents with the FollowUpRequired flag set.

Alternatively, the “Count” operator can be used to query the number of completed incidents using filtering options. The following are URI examples:

```
http://localhost:8800/Oms/CompletedIncidents/$count  
http://localhost:8800/oms/CompletedIncidents/$count/?$filter=FollowupRequired eq true
```

6. Navigation to External Maps

The interface defined in sections below allows for a selected DNOM object (i.e., vault or line on the Network View) the following actions:

- Navigating to a constructed latitude/longitude-based URL to Google Maps or another external map service.
- Locating the object on these maps.

6.1. Defining Interface

The following interface needs to be implemented:

```
public interface ICoordinatesCovertor
{
    void GetURI(string inputString, ref string URI);
    void ToLatitudeLongitude(ref double Latitude, ref double Longitude, double X, double Y);
    void ToXY(double Latitude, double Longitude, ref double X, ref double Y);
}
```

A sample below illustrates the interface implementation:

```
public class CoordinatesCovertor : ICoordinatesCovertor
{
    #region ICoordinatesCovertor Members

    /// <summary>
    /// This method generate an URI to external map based on the input string
    /// </summary>
    /// <param name="inputString">input string contains all the info needed to generate an URI to
    external map</param>

    public void GetURI(string inputString, ref string URI)
    {
        URI = @"https://maps.google.com/maps?q=24.197611,120.780512&z=8";
    }

    /// This method converts x, y to latitude, longitude.
    public void ToLatitudeLongitude(ref double Latitude, ref double Longitude, double X, double Y)
    {
        Latitude = X;
        Longitude = Y;
    }

    /// This method converts latitude, longitude to x, y.
    public void ToXY(double Latitude, double Longitude, ref double X, ref double Y)
    {
        X = Latitude;
        Y = Longitude;
    }
}
```

```
}  
#endregion  
}
```

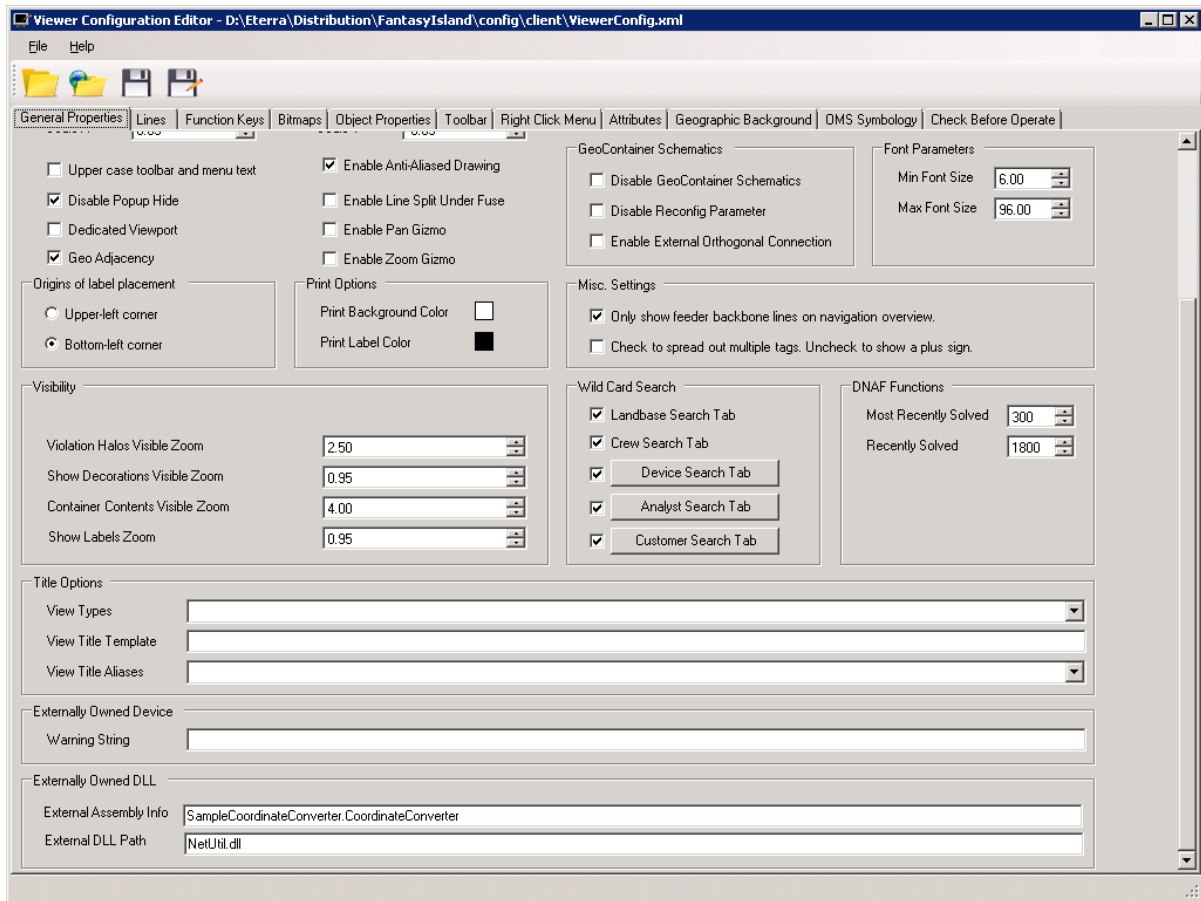
6.2. Configuring External Assembly

Before using an external assembly implementing the ICoordinatesConverter interface, you need to configure the following parameters in the Viewer Configuration Editor tool:

- External Assembly Info
- External DLL Path

To configure parameters:

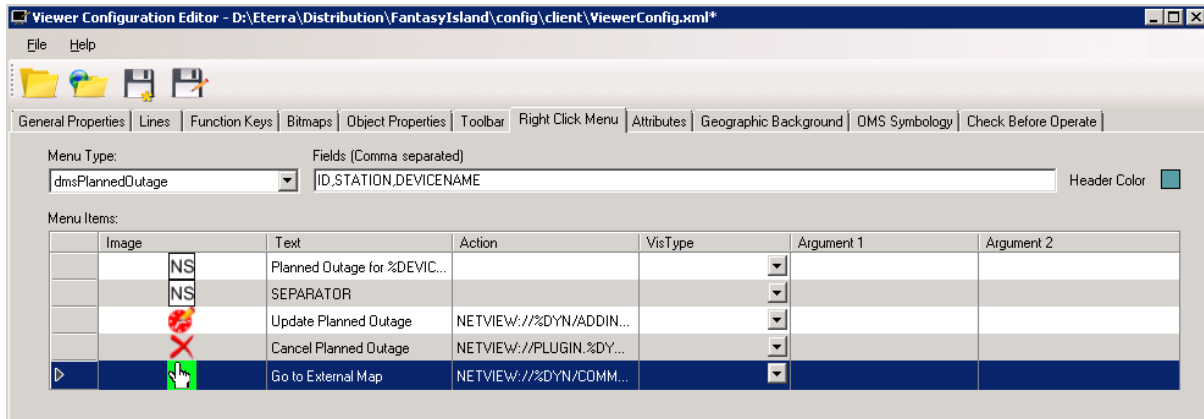
1. Open the Viewer Configuration Editor tool and navigate to the General Properties tab.
2. In the Externally Owned DLL:
 - Enter the external assembly name and type into the External Assembly Info field using the following format:
AssemblyName.TypeName
 - Enter the full path to an external DLL file in the External DLL Path field.
3. Save changes via the File >Save menu.



6.3. Creating Navigation Command

The steps below describe how to create a right-click command to navigate to and locate devices on an external map.

1. Open the Viewer Configuration Editor tool and navigate to the Right-Click Menu tab.
2. Select the menu type (for example, dmsPlannedOutage).
3. In the Menu Items grid, click Insert to add a new menu item.
4. Enter the following information:
 - **Text:** Go to External Map
 - **Action:** NETVIEW://%DYN/COMMAND/NAVIGATEEXTERNALMAP/?X=%XINT&Y=%YINT
5. Save changes via the File > Save menu.



6.4. Defining a GetURL Method

Use the following format for the inputString parameter:

```
Latitude=12,Longitude=34
```


7. OMS Query Web Service

Together with the common Net Data Server web service, Net Data Server exposes the OMS Query web service (OmsQueryService), which is a generic web API that supports OMS queries.

The OMS Query web service is used by the Data Correction Interface application, which supplies historical order and incident data to the Data Correction grid tabular display of the OMS Display plugin for data review and correction. For more information, refer to the *Enterprise OMS Data Correction Interface User Guide*.

OmsQueryService exposes a simple endpoint for OMS queries using the “select”, “from”, and “where” query syntax. The OMS Query web service has the following URL:

```
http://localhost:8443/api/Oms/v2/Oms
```

Request format:

```
http://localhost:8443/api/Oms/v2/Oms?select=<Query fields from the Incident object>&from=<Queried collection>&where=<Conditions>
```

Example request:

```
http://localhost:8443/api/Oms/v2/Oms?select=Id,Cmi,PlannedOutageId,CallCount,IsOverhead,Station,ServiceCenter,LocationDescription,State,County,Municipality,Division,Cleared,OpenDevicePtX,OpenDevicePtY,Name,SeqNumber,IncidentStatus&from=OMS.RT.COMPLETEDINCIDENT&where=id=6c167e6d-d00b-472a-8b1c-ba098bd821a5
```

Example response:

```
[
  {
    "Id": "6c167e6d-d00b-472a-8b1c-ba098bd821a5",
    "Cmi": 0,
    "PlannedOutageId": "",
    "CallCount": 0,
    "IsOverhead": false,
    "Station": "RAVEN",
    "ServiceCenter": "3LIONSSC",
    "LocationDescription": "Pole# 141631273; 8286 West St, Roarke",
    "State": "Washington",
    "County": "Gordon",
    "Municipality": "Roarke",
    "Division": "ROARKE",
    "Cleared": false,
    "OpenDevicePtX": 936718.4,
    "OpenDevicePtY": 811147.2,
    "Name": "2780000005",
    "SeqNumber": 5,
    "IncidentStatus": 7
  }
]
```

]